

Parity equidistribution of nested floor powers, with descent applications to the Juggler map

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Abstract

For odd n set $m = \lfloor n^{3/2} \rfloor$, $v = \lfloor m^{3/2} \rfloor$, and so on: the integer chains obtained by iterating $x \mapsto \lfloor x^{3/2} \rfloor$ and $x \mapsto \lfloor \sqrt{x} \rfloor$ in a prescribed pattern. We prove parity equidistribution, with power savings, for every pattern of depth at most four, and for selected deeper patterns. The obstacle at depth two and beyond is that the naive expansion of $m^{3/2}$ leaves the sawtooth $\{n^{3/2}\}$ with an amplitude that grows like $n^{3/4}$, which defeats the classical van der Corput method. Two devices remove it. An *exact linearization* kills the inner floor identically — $m^{3/2} = \frac{3}{2}mn^{3/4} - \frac{1}{2}n^{9/4} + O(n^{-3/4})$ with a one-signed remainder — so that the non-smooth integer m enters the phase linearly with a smooth coefficient. Deeper patterns then funnel into a single *kernel*: the exponential sum of the level-2 floor defect $\{\lfloor n^{3/2} \rfloor^{3/2}\}$ against smooth weights of scale $n^{9/8}$. The central result (Theorem 5.3) is a power-saving bound for that kernel, $K_c \ll P^{1-1/96+\varepsilon}$, by double Weyl differencing over an exact carry-branch decomposition and master identity, with a targeted third differencing for the exact level-2 wave pieces (Lemma 5.2).

The sequences arise as the itineraries of the Juggler map $J(n) = \lfloor \sqrt{n} \rfloor$ (n even), $\lfloor n^{3/2} \rfloor$ (n odd), whose finite exact theory is developed in a companion manuscript [22]. As corollaries, the classes of starts carrying a uniform power-envelope descent certificate of length at most four, five, seven,

and eight have natural densities $13/16$, $7/8$, $57/64$, and $29/32$; an unconditional counting argument shows that parity equidistribution at *all* depths would give density one to the set of starts with some finite descent certificate. None of these is a density of starts that reach 1, and no statement about the Juggler conjecture itself is claimed. The remaining obstacle is the level-3 kernel, where the weight scale $n^{27/16}$ exceeds n ; we state it as an open problem, prove an almost-every-shift form of its model, and record precisely what blocks the deterministic case.

1. Introduction

Equidistribution questions for a *single* floor of a smooth function — the Piatetski-Shapiro tradition — are classical and well developed. This paper concerns the *composition* of floors:

$$m = \lfloor n^{3/2} \rfloor, \quad v = \lfloor m^{3/2} \rfloor, \quad w = \lfloor v^{3/2} \rfloor, \quad z = \lfloor w^{3/2} \rfloor, \dots$$

and, specifically, the joint distribution of the *parities* along such chains. Fix a pattern (an *itinerary word*) $w_1 w_2 \dots w_d$ over the alphabet $\{E, O\}$, where an O applies $x \mapsto \lfloor x^{3/2} \rfloor$ and an E applies $x \mapsto \lfloor \sqrt{x} \rfloor$. The question is whether the 2^d parity classes of the resulting chain equidistribute over odd starts $n \leq N$, with a power saving.

The difficulty is genuinely one of composition. Writing $\theta = \{n^{3/2}\}$, the naive expansion $m^{3/2} = n^{9/4} - \frac{3}{2}\theta n^{3/4} + O(n^{-3/4})$ leaves a sawtooth of *growing* amplitude $n^{3/4}$ inside the phase. No estimate for exponential sums with smooth monomial phases applies; the phase is not smooth, and the sawtooth cannot be discarded. All results in the single-floor literature stop before this point (Section 2).

Two ideas carry the paper.

1. **Exact linearization** (Lemma 4.3). The identity $m^{3/2} = \frac{3}{2}mn^{3/4} - \frac{1}{2}n^{9/4} + E(n)$ with $0 \leq E(n) \leq \frac{1}{2}n^{-3/4}$ eliminates the inner floor *exactly*: the integer m enters linearly, multiplied by a smooth coefficient, and the only non-smooth object left is m itself, which is handled by differencing and by an exact gap-cell decomposition. This device, iterated along the chain with one-signed Taylor remainders at every level, is what makes depth ≥ 2 accessible.
2. **The kernel theorem** (Theorem 5.3). Every reorganization of the depth-4 pattern OOO^* funnels into one object, the exponential sum of the level-2 floor defect

$$K_c(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e(c(n) \{ \lfloor n^{3/2} \rfloor^{3/2} \}), \quad c \asymp kP^{9/8}.$$

We prove $K_c(P) \ll P^{1-1/96+\varepsilon}$, uniformly for $k \leq P^{1/24}$, by double Weyl differencing over an exact carry-branch decomposition and master identity (Lemma 5.1), with a targeted third differencing for the level-2 wave pieces (Lemma 5.2). This is the hardest result of the paper, and Section 5 is written at full length — every estimate displayed with its constant — so that it can be checked without reference to anything outside this manuscript.

With the kernel theorem, depth-4 parity equidistribution is complete (Theorem 6.1), and the same engine counts selected contracting patterns of lengths five, seven, and eight (Theorems 6.2–6.4).

The chains are not chosen at random: they are the itineraries of the Juggler map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even}, \\ \lfloor n^{3/2} \rfloor, & n \text{ odd}, \end{cases}$$

introduced by Pickover [1] (OEIS A094683 [2]); universal arrival at 1 is open. The map is niche; the nested-floor sums must stand on their own, and the paper is organized so that they do. The exact finite-word calculus of J — the power envelope $J^{|w|}(n)^{2^{|w|}} \leq n^{3^{\#O(w)}}$, the defect identity, and a small-cycle census — is a companion manuscript [22]; here we import only the contraction criterion (Proposition 3.1). The dynamical payoff of the counting theorems is a sequence of *certified-descent densities*: the set of starts guaranteed to drop below their starting value within four, five, seven, eight steps has natural density $13/16$, $7/8$, $57/64$, $29/32$ (Corollaries 4.9 and 6.5), and equidistribution at all depths would give the set of starts with *some* finite descent certificate density one (Proposition 7.1). We state plainly what these corollaries are not: they are not densities of starts that reach 1, and they do not touch the Juggler conjecture, whose analogue of Terras’s almost-all theorem for Collatz [4, 5, 6] remains open.

Section 7 states the frontier precisely. One nesting deeper, the same kernel reappears with weight scale $n^{27/16} > n$ (Conjecture 7.3), and every method of this paper stops below that scale. What survives is a clean model problem — the amplitude-product sums $\sum e(A(t)\{B(t)\})$ with $1 \ll A' \ll A$ — for which we prove two-sided square-root cancellation for almost every shift of the fractional argument (Theorem 7.4), by direct integration and with no harmonic analysis; the deterministic instance (Conjecture 7.5) is open, and we record in three sentences why the obvious shortcuts fail.

1.1 Verification and evidence conventions

Every theorem in this paper is an ordinary human proof from the stated classical inequalities; none is machine-checked. The exact floor identities beneath the analysis — the parity bridge $\lfloor x \rfloor$ odd iff $\{x/2\} \geq 1/2$, the gap-cell identity, and the double-gap identity used by the kernel theorem — are formalized in Lean in the companion repository (`floor_odd_iff_half_le_fract_half`, `floor_gap_eq_carry`, `seq_floor_gap`, `seq_floor_gap_second`), as are the finite-word theorems of the companion manuscript [22]. Scaled-integer validations of the linearization identities exist in the same repository; they are checks, not proofs, and no numerical computation is used as a step in any proof below. Densities of certificate classes are never densities of starts that reach 1.

2. Related work

Single floors. The distribution of $\lfloor n^c \rfloor$ ($c > 1$, $c \notin \mathbb{N}$) in arithmetic progressions and among primes goes back to Piatetski–Shapiro [12]; Leitmann [13] treated general smooth $\lfloor f(n) \rfloor$, and the prime exponent for $\lfloor n^c \rfloor$ has a long refinement history, e.g. Rivat–Sargos [14] and Rivat–Wu [15]. All of these concern a single floor of a smooth function; the analytic core is an exponential sum with smooth monomial phase, estimated by van der Corput’s method in the form presented by Graham–Kolesnik [11].

Digital and parity properties of $\lfloor n^c \rfloor$. Mauduit–Rivat [16] established q -multiplicative equidistribution along $\lfloor n^c \rfloor$ for c in an explicit range, and Morgenbesser [17] proved the analogous statements for the sum of digits of $\lfloor n^c \rfloor$ in residue classes. Parity of $\lfloor n^{3/2} \rfloor$ — our depth-1 statement, Theorem 4.1 — is the simplest instance of this circle and is classical; we include the short proof to fix constants and notation.

Arithmetic of Piatetski–Shapiro sequences. Baker, Banks, Brüdern, Shparlinski, and Weingartner [18] study squarefree values, prime factors, and Carmichael numbers along $\lfloor n^c \rfloor$; Glasscock [21] studies linear equations whose unknowns range over Piatetski–Shapiro sequences. These results intersect a Piatetski–Shapiro sequence with other arithmetic sets; none composes two floors.

Beatty sequences. For the linear case $\lfloor \alpha n \rfloor$ the discrepancy theory is governed by the continued fraction of α (three-distance/Ostrowski structure); see Abercrombie [19] and, for character sums along Beatty sequences, Banks–Shparlinski [20]. The linear case admits exact self-similar structure that the convex case $n^{3/2}$ does not.

What is not covered. We know of no published equidistribution or parity result for *nested* floor powers $\lfloor \lfloor n^c \rfloor^d \rfloor$ with $c, d > 1$, nor any treatment of the level-2 defect sums $K_c(P)$ of Section 5. The obstruction is structural, not incremental: after one floor the argument of the second floor is an integer sequence, not a smooth function, and its fractional defect enters later phases with growing amplitude. The exact-linearization device of Lemma 4.3 and the carry-branch decomposition of Lemma 5.1 are, to our knowledge, new. A literature check was last refreshed in August 2026; we would welcome pointers to anything missed.

Dynamical context. For the Collatz map, Terras [4] and Everett [5] proved that almost every start has finite stopping time, and Tao [6] proved that almost all orbits attain almost bounded values. These are methodological cousins — density statements through parity-word counting — and motivate Proposition 7.1; they prove nothing about J , whose branches are floor powers rather than affine maps. Prasad–Prasad [7] estimate excursion constants for juggler-like random walks; we use that comparison only descriptively.

3. Preliminaries

Throughout, n denotes an odd integer, $e(t) = e^{2\pi it}$, $\{x\}$ the fractional part, and $\psi(x) = (-1)^{\lfloor x \rfloor}$. On a dyadic block $n \in (P, 2P]$ we write $n \sim P$. For the chains,

$$X = n^{3/2}, \quad m = \lfloor X \rfloor, \quad \theta = X - m; \quad Y = m^{3/2}, \quad v = \lfloor Y \rfloor, \quad \theta_2 = Y - v;$$

and on even m , $U = m^{1/2}$, $w = \lfloor U \rfloor$, $\theta_w = U - w$. An itinerary word of depth d at n is the sequence of parities $\text{word}_d(n)$ of $n, J(n), \dots, J^{d-1}(n)$.

Proposition 3.1 (power-envelope contraction; companion [22]). If the word w is realized at $n \geq 2$ (the orbit’s parities are the letters of w) and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Induction along the word gives $J^{|w|}(n)^{2^{|w|}} \leq n^{3^{\#O(w)}}$: an even letter squares to at most the state, an odd letter squares to at most the cube. The exponent gap and $n \geq 2$ force $J^{|w|}(n)^{2^{|w|}} < n^{2^{|w|}}$. (Lean: `power_bound_word`, `power_bound_contracts` in the companion repository.) $\square$

A word w with $3^{\#O(w)} < 2^{|w|}$ is *contracting*; realizing a contracting word is a *descent certificate* of length $|w|$. Contracting words used below: E , OE , $OOEE$, $OOOEE$, $OOEOE$, $OOEOOEE$, $OOOEEOEE$, and the length-8 quartet of Theorem 6.4. Every even start realizes E , and every odd start with even image realizes OE ; those two certificates cover all starts except the odd-to-odd class, which is where the counting problem lives.

Lemma 3.2 (parity bridge). $\lfloor x \rfloor$ is odd if and only if $\{x/2\} \geq \frac{1}{2}$. (Lean: `floor_odd_iff_half_le_fract_half`.)

Proof. Write $x = \lfloor x \rfloor + \{x\}$ and split by the parity of $\lfloor x \rfloor$. $\square$

Lemma 3.3 (van der Corput, second-derivative form [11]). Let f be twice differentiable on an interval of length M , with $\lambda \leq |f''| \leq \alpha\lambda$ for some $\alpha \geq 1$. Then

$$\left| \sum e(f) \right| \ll_{\alpha} M\lambda^{1/2} + \lambda^{-1/2},$$

the sum over the integers of the interval.

Lemma 3.4 (Erdős–Turán [8]). The discrepancy of R points $x_j \in \mathbb{R}/\mathbb{Z}$ against an interval satisfies, for every $H \geq 1$,

$$D \ll \frac{R}{H} + \sum_{h=1}^H \frac{1}{h} \left| \sum_{j \leq R} e(hx_j) \right|.$$

Lemma 3.5 (Vaaler [10]). For every $J \geq 1$ there are trigonometric polynomials $V_J(t) = \sum_{0 < |q| \leq J} a_q e(qt)$ with $|a_q| \leq \min(1, \frac{2}{|q|})$ and a nonnegative trigonometric polynomial $\Delta_J \geq 0$ of degree J with constant term and coefficients $O(1/J)$, such that the period-2 square wave satisfies $\psi(x) = V_J(x/2) + O(\Delta_J(x/2))$. For products, $|\psi_1 \psi_2 - V^{(1)} V^{(2)}| \leq \Delta^{(1)} + \Delta^{(2)} + \Delta^{(1)} \Delta^{(2)}$, and each error term is again a nonnegative trigonometric polynomial of the same shape.

The class indicators below are products of half-wave factors $\frac{1}{2}(1 \pm \psi)$ evaluated along *formal* chains — chains whose branches are dictated by the target word, not by the orbit. The next lemma is the exact identity that justifies this; it replaces any appeal to sampled verification.

Lemma 3.6 (branch consistency). Let $d \geq 1$ and let $w = w_1 \cdots w_d$ be a word with $w_1 = O$. For odd n , define the formal chain $x_1 = n$ and, for $1 \leq t < d$,

$$\xi_t = \begin{cases} x_t^{3/2}, & w_t = O, \\ x_t^{1/2}, & w_t = E, \end{cases} \quad x_{t+1} = \lfloor \xi_t \rfloor,$$

and set $\sigma_t = +1$ if $w_t = E$, $\sigma_t = -1$ if $w_t = O$. Then, for every odd n ,

$$\prod_{t=1}^{d-1} \frac{1}{2} (1 + \sigma_{t+1} \psi(\xi_t)) = [\text{word}_d(n) = w].$$

Proof. Induct on d . For $d = 1$ both sides are 1. Let Π_{d-1} be the product over $t \leq d-2$; by induction $\Pi_{d-1} = [\text{word}_{d-1}(n) = w_1 \cdots w_{d-1}]$. If $\Pi_{d-1} = 0$, both sides vanish, since every factor lies in $[0, 1]$. If $\Pi_{d-1} = 1$, then the orbit realizes $w_1 \cdots w_{d-1}$, and a second induction along the chain gives $x_t = J^{t-1}(n)$ for $t \leq d-1$: assuming $x_t = J^{t-1}(n)$, the true parity of x_t is w_t , so J applies exactly the w_t -branch and $J^t(n) = \lfloor \xi_t \rfloor = x_{t+1}$. Hence ξ_{d-1} is the true real image with $\lfloor \xi_{d-1} \rfloor = J^{d-1}(n)$, and the last factor $\frac{1}{2}(1 + \sigma_d \psi(\xi_{d-1}))$ equals 1 exactly when the parity of $J^{d-1}(n)$ is w_d , else 0. $\square$

Each $\psi(\xi_t)$ is defined for *every* odd n , which is what permits the unconditional Vaaler expansion of the product: off the cylinder, the vanishing of an earlier factor kills the term, and on the cylinder every factor computes the true letter.

The last three preliminaries are the remaining analytic tools of Section 5: the third-derivative test, a finite Fourier expansion for sawtooth phases whose coefficient may exceed 1, and a second/third-derivative test for two-term monomial phases that is applied where two curvature scales can collide.

Lemma 3.7 (van der Corput, third-derivative form [11, Thm. 2.6]). Let f be three times differentiable on an interval of length M , with $\lambda_3 \leq |f'''| \leq \alpha \lambda_3$. Then

$$\left| \sum e(f) \right| \ll_{\alpha} M \lambda_3^{1/6} + M^{1/2} \lambda_3^{-1/6},$$

the sum over the integers of the interval.

Lemma 3.8 (finite Fourier expansion with a shifted window). Let $B \in \mathbb{R}$, and let T, J be integers with $J \geq 1$ and $T \geq 8(1 + |B|)$. There exist coefficients $(b_u)_{|u| \leq T}$ with

$$|b_u| \leq \min\left(2, \frac{1}{\pi|u+B|}\right) + \min\left(2, \frac{1}{\pi|u|}\right), \quad \sum_{|u| \leq T} |b_u| \ll \log(2 + |B|),$$

and coefficients $(v_q)_{0 < |q| \leq J}$ with $|v_q| \leq \min(2, 2\pi|B|)/|q|$, such that for every $t \in \mathbb{R}$

$$\left| e(-B\{t\}) - \sum_{|u| \leq T} b_u e(ut) - \sum_{0 < |q| \leq J} v_q e(qt) \right| \leq \min(2, 2\pi|B|) \Delta_J(t) + \frac{8(1 + |B|)}{T},$$

where $\Delta_J \geq 0$ is a trigonometric polynomial of degree J with constant term and coefficients at most $1/(J+1)$.

Proof. Write $f(t) = e(-B\{t\})$ and $\sigma(t) = \frac{1}{2} - \{t\}$, the sawtooth with jump $+1$ at the integers. The jump of f at the integers is $1 - e(-B)$, so $g := f - (1 - e(-B))\sigma$ is continuous, 1-periodic, and piecewise C^1 . Its Fourier coefficients are, for $u \neq 0$,

$$\hat{g}(u) = \frac{1 - e(-B)}{2\pi i} \left(\frac{1}{u+B} - \frac{1}{u} \right) = -\frac{(1 - e(-B))B}{2\pi i u(u+B)}, \quad |\hat{g}(u)| \leq \frac{\min(2, 2\pi|B|)}{2\pi} \frac{|B|}{|u||u+B|},$$

together with $|\hat{g}(0)| \leq 1$; in particular $|\hat{g}(u)| \leq \min(2, \frac{1}{\pi|u+B|}) + \min(2, \frac{1}{\pi|u|})$ and, by partial fractions, $\sum_u |\hat{g}(u)| \ll \log(2 + |B|)$: the series converges absolutely, so g equals its Fourier series everywhere, and the tail beyond $|u| \leq T \geq 8(1 + |B|)$ satisfies (using $|u+B| \geq |u|/2$ there)

$$\sum_{|u| > T} |\hat{g}(u)| \leq \frac{2|B|}{\pi} \sum_{|u| > T} \frac{1}{u^2} \cdot 2 \leq \frac{8|B|}{\pi T} \leq \frac{8(1 + |B|)}{T}.$$

Take $b_u = \hat{g}(u)$. For the sawtooth part, Vaaler's theorem in its original form [10] provides a trigonometric polynomial $V_J^*(t) = \sum_{0 < |q| \leq J} a_q^* e(qt)$ with $|a_q^*| \ll 1/|q|$ and $|\sigma(t) - V_J^*(t)| \leq \Delta_J(t)$ with Δ_J as stated; multiply by $1 - e(-B)$, whose modulus is $2|\sin \pi B| \leq \min(2, 2\pi|B|)$, and set $v_q = (1 - e(-B))a_q^*$. $\square$

When the lemma is applied with $t = G(n)$ along a block, the Δ_J -term is a nonnegative majorant: its sum over the block is $P/(J+1)$ plus J -truncated mode sums of G with coefficients $\leq 1/(J+1)$, and the flat term contributes $8(1 + |B|)P/T$. Both costs are displayed at each application below.

Lemma 3.9 (two-term monomial test). Let $E \subset \mathbb{Q} \cap [-4, 4]$ be a fixed finite set disjoint from $\{0, 1, 2, 3\}$, let $\alpha \neq \beta$ lie in E , let $I \subseteq (P, 2P]$, and let

$$f(n) = a n^\alpha + b n^\beta + g(n), \quad M := \max(|a|P^{\alpha-2}, |b|P^{\beta-2}) \leq 1, \quad ab \neq 0,$$

where the perturbation satisfies, for every $n \in I$,

$$|g''(n)| \leq \rho (|a|n^{\alpha-2} + |b|n^{\beta-2}), \quad |g'''(n)| \leq \rho (|a|n^{\alpha-3} + |b|n^{\beta-3}),$$

with $\rho \leq \rho_0(E)$ sufficiently small in terms of E alone. Then

$$\left| \sum_{n \in I} e(f(n)) \right| \ll_E |I| M^{1/2} + M^{-1/2} + |I| (M/P)^{1/6} + |I|^{1/2} (M/P)^{-1/6}.$$

Proof. Write $A(n) = a\alpha(\alpha-1)n^{\alpha-2}$ and $B(n) = b\beta(\beta-1)n^{\beta-2}$, so that

$$f'' = A + B + g'', \quad n f''' = (\alpha-2)A + (\beta-2)B + n g'''.$$

On the block, $|A(n)| \asymp_E |a|P^{\alpha-2}$ and $|B(n)| \asymp_E |b|P^{\beta-2}$. The ratio $|A/B| = |a\alpha(\alpha-1)/(b\beta(\beta-1))| n^{\alpha-\beta}$ is strictly monotone in n , so for any threshold $K = K(E) \geq 3$ the block splits into at most three consecutive intervals: I_1 where $|A/B| \geq K$, I_2 where $|A/B| \leq 1/K$, and a middle I_0 .

On I_1 : $|f''| \geq (1 - \frac{1}{K})|A| - |g''| \geq \frac{1}{2}|A| \gg_E M$ -scale, with $\sup |f''|/\inf |f''| \leq C(E)$; Lemma 3.3 gives $\ll_E |I|M^{1/2} + M^{-1/2}$. Symmetrically on I_2 .

On I_0 , $|A| \asymp_K |B|$. If A and B have the same sign there, $|f''| \geq |A| + |B| - |g''| \geq |A|$ and Lemma 3.3 applies as before. If they have opposite signs, write $B = -sA$ with $s = s(n) \in [1/K, K]$, monotone in n . Then

$$f'' = A(1 - s) + g'', \quad n f''' = A((\alpha - 2) - s(\beta - 2)) + n g'''.$$

The two affine functions $s \mapsto 1 - s$ and $s \mapsto (\alpha - 2) - s(\beta - 2)$ have distinct zeros: $s = 1$ and $s = (\alpha - 2)/(\beta - 2) \neq 1$ (here $\alpha \neq \beta$ and $\beta \neq 2$ are used). Hence

$$c_6(E) := \min_{s>0} \max(|1 - s|, |(\alpha - 2) - s(\beta - 2)|) > 0,$$

so at every $n \in I_0$ either $|f''| \geq \frac{1}{2}c_6|A|$ or $|n f'''| \geq \frac{1}{2}c_6|A|$, after absorbing g by $\rho_0(E) \leq c_6/8$. Since $s(n)$ is monotone, the set where $|1 - s(n)| \geq c_6$ is the complement of a single interval, so I_0 splits into at most three consecutive intervals, each either entirely second-derivative-good (Lemma 3.3 at scale $\gg_E M$ -normalized: $\ll |I|M^{1/2} + M^{-1/2}$) or entirely third-derivative-good (Lemma 3.7 at scale $|f'''| \gg_E M/P$ -normalized: $\ll |I|(M/P)^{1/6} + |I|^{1/2}(M/P)^{-1/6}$). Summing the $O_E(1)$ contributions proves the lemma. $\square$

The proof's only content is that the second and third derivatives of a two-term monomial phase cannot both be small on the same subinterval — the linear system with matrix $(\begin{smallmatrix} 1 & 1 \\ \alpha-2 & \beta-2 \end{smallmatrix})$ is invertible — so the interval splits into boundedly many pieces on which one of the two classical tests runs at full scale. All applications in Section 5 use exponent pairs from $E = \{\frac{3}{4}, \frac{11}{8}, \frac{3}{2}, \frac{15}{8}\}$. One corner of Section 5 (three curvature scales meeting on a zero-offset branch) needs the three-term extension, with the fourth-derivative test as the final fallback.

Lemma 3.10 (three-term monomial test). Let E be as in Lemma 3.9 and let $\alpha, \beta, \gamma \in E$ be pairwise distinct, $I \subseteq (P, 2P]$, and

$$f(n) = a n^\alpha + b n^\beta + c n^\gamma + g(n), \quad M := \max(|a|P^{\alpha-2}, |b|P^{\beta-2}, |c|P^{\gamma-2}) \leq 1,$$

with $abc \neq 0$ and g satisfying the analogues of the perturbation bounds of Lemma 3.9 for the second, third, and fourth derivatives, at some $\rho \leq \rho_0(E)$. Then

$$\left| \sum_{n \in I} e(f(n)) \right| \ll_E |I| M^{1/2} + M^{-1/2} + |I| (M/P)^{1/6} + |I|^{1/2} (M/P)^{-1/6} + |I| (M/P^2)^{1/14} + |I|^{3/4} (M/P^2)^{-1/14}.$$

Proof. As in Lemma 3.9, with the 2×2 system replaced by the 3×3 system

$$f'' = A + B + C + g'', \quad n f''' = (\alpha-2)A + (\beta-2)B + (\gamma-2)C + n g''',$$

$$n^2 f'''' = (\alpha-2)(\alpha-3)A + (\beta-2)(\beta-3)B + (\gamma-2)(\gamma-3)C + n^2 g'''' ,$$

where A, B, C are the three curvature terms. The coefficient matrix is the Vandermonde-type matrix in the distinct values $\alpha-2, \beta-2, \gamma-2$ (rows 1, x , $x(x-1)$), hence invertible with inverse bounded in terms of E : at every point, $\max(|f''|, |n f'''|, |n^2 f''''|) \geq c_7(E) \max(|A|, |B|, |C|) \geq c_7(E) \tilde{M}$, $\tilde{M} \asymp_E M$ -normalized, after absorbing g by $\rho_0 \leq c_7/8$. The ratios A/B , B/C , A/C are monotone, so I splits into $O_E(1)$ consecutive intervals on each of which one fixed derivative order $r \in \{2, 3, 4\}$ is good at full scale; Lemma 3.3 ($r = 2$), Lemma 3.7 ($r = 3$), or the fourth-derivative test [11, Thm. 2.6, $k = 4$] ($\ll |I|\lambda_4^{1/14} + |I|^{3/4}\lambda_4^{-1/14}$ at $\lambda_4 \asymp_E M/P^2$) finishes each piece. $\square$

4. Exact linearization and depths one to four

For odd n write $s(n) = \psi(n^{3/2}) = (-1)^m$ and $S_O(N) = \sum_{n \leq N, n \text{ odd}} s(n)$. Let $M(N)$ be the number of odd $n \leq N$ and $\text{OO}(N) = \#\{n \leq N : n \text{ odd}, J(n) \text{ odd}\}$, so $S_O(N) = M(N) - 2 \text{OO}(N)$. By Lemma 3.2, with $g(r) = \frac{1}{2}(2r+1)^{3/2}$, $s(2r+1) = -1$ iff $\{g(r)\} \geq \frac{1}{2}$, and $S_O(N)$ is (twice) an interval discrepancy of the sequence $\{g(r)\}$.

Theorem 4.1 (depth one; classical). $|S_O(N)| \ll N^{5/6}$.

Proof. This is the standard van der Corput estimate for the monomial g , included for completeness. $g''(r) = \frac{3}{2}(2r+1)^{-1/2}$ is positive and decreasing. Partition $1 \leq r < R$ into dyadic blocks $[M, \min(2M, R))$; on a block $g'' \asymp M^{-1/2}$, so for the h -th mode $f = hg$ has $\lambda \asymp hM^{-1/2}$ and Lemma 3.3 gives $|\sum_{r \asymp M} e(hg(r))| \ll h^{1/2}M^{3/4} + h^{-1/2}M^{1/4}$. Lemma 3.4 with $H = \lfloor M^{1/6} \rfloor$ bounds the block's discrepancy by $\ll M^{5/6}$, and the dyadic sum is $O(R^{5/6}) = O(N^{5/6})$. $\square$

Corollary 4.2 (density of the odd-to-odd class). $|\text{OO}(N) - N/4| \ll N^{5/6}$. Equivalently, the starts covered by the uniform one- or two-step certificates (E and OE) have natural density $3/4$.

This is a density of a certificate class, not of starts that reach 1. Theorem 4.1 concerns consecutive source intervals; it supplies no transfer theorem for orbit samples or sparse image sets, so it cannot by itself control the odd-to-odd dynamics after the first step. The rest of the paper does not transfer it; it attacks the nested sums directly, by removing the inner floors exactly before any analytic step.

Lemma 4.3 (exact linearization and gap cells). (i) For every odd $n \geq 3$,

$$m^{3/2} = \frac{3}{2} m n^{3/4} - \frac{1}{2} n^{9/4} + E(n), \quad 0 \leq E(n) \leq \frac{3}{8} (n^{3/2} - 1)^{-1/2} \leq \frac{1}{2} n^{-3/4}.$$

(ii) For $h \geq 1$ and odd n , let $g(n) = m(n+2h) - m(n)$ and $\delta(n) = (n+2h)^{3/2} - n^{3/2}$. Then

$$g(n) = \lfloor \delta(n) \rfloor + \kappa(n), \quad \kappa(n) = [\{n^{3/2}\} \geq 1 - \{\delta(n)\}] \in \{0, 1\}.$$

Proof. (i) Let $f(t) = (X-t)^{3/2}$ on $[0, \theta]$, so $f(\theta) = m^{3/2}$. Taylor with Lagrange remainder at 0 gives $f(\theta) = X^{3/2} - \frac{3}{2}X^{1/2}\theta + \frac{3}{8}(X-\xi)^{-1/2}\theta^2$ for some $\xi \in (0, \theta)$. Substituting $\theta = X - m$ in the linear term yields $-\frac{1}{2}X^{3/2} + \frac{3}{2}mX^{1/2} = -\frac{1}{2}n^{9/4} + \frac{3}{2}mn^{3/4}$, and the remainder lies in $[0, \frac{3}{8}(X-1)^{-1/2}]$, with $(X-1)^{-1/2} \leq \frac{4}{3}X^{-1/2}$ already for $n \geq 3$. (ii) $g = \lfloor X + \delta \rfloor - \lfloor X \rfloor = \lfloor \delta \rfloor + \lfloor \{X\} + \{\delta\} \rfloor$, and the last floor is 1 precisely when $\{X\} + \{\delta\} \geq 1$ (Lean: `floor_gap_eq_carry`, `seq_floor_gap`). $\square$

The point of (i): the non-smooth integer m enters the phase *linearly*, multiplied by a smooth coefficient; no fractional part of growing amplitude survives. The naive expansion $m^{3/2} = n^{9/4} - \frac{3}{2}\theta n^{3/4} + O(n^{-3/4})$ instead leaves the sawtooth θ with the growing amplitude $n^{3/4}$, which defeats the van der Corput method directly. For (ii), since δ is smooth and increasing with $\delta' \asymp hn^{-1/2}$, the level sets of $\lfloor \delta \rfloor$ partition a dyadic block $n \in (P, 2P]$ into $O(hP^{1/2})$ cells of length $\asymp P^{1/2}/h$ on which the integer part of the gap is constant.

Theorem 4.4 (nested parity discrepancy). For every $\varepsilon > 0$ and every $(a, b) \in \{0, 1\}^2 \setminus \{(0, 0)\}$,

$$\left| \sum_{n \leq N, n \text{ odd}} \psi(n^{3/2})^a \psi(m(n)^{3/2})^b \right| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Consequently each of the four joint parity classes of $(m, \lfloor m^{3/2} \rfloor)$ on odd $n \leq N$ has cardinality $\frac{N}{8} + O(N^{23/24+\varepsilon})$; in particular the odd-to-odd cylinder refines one letter deeper than Theorem 4.1.

Proof. Work on dyadic blocks $n \in (P, 2P]$, odd, with truncations $J_1 = J_2 = P^{1/24}$ (wave modes), $H = P^{1/12}$ (differencing), and $R = P^{1/4}$ (cell modes).

Step 1 (wave expansion). By Lemma 3.5 applied to each factor, expanding both waves reduces the theorem to bounds, uniform in the mode pair, for

$$S_{\mu,\nu}(P) = \sum_{n \in (P, 2P], n \text{ odd}} e(\mu n^{3/2} + \nu m^{3/2}), \quad |\mu| \leq J_1, \quad \frac{1}{2} \leq |\nu| \leq J_2,$$

with $\mu, \nu \in \frac{1}{2}\mathbb{Z}$; the pure first-factor majorant sums are $\ll P/J_1 + P^{5/6+\varepsilon} \ll P^{23/24+\varepsilon}$ by the depth-1 machinery, and mode weights sum to $O(\log^2 P)$. Write $i = 2\mu$, $j = 2\nu$.

Step 2 (linearization). By Lemma 4.3(i), replacing $\frac{j}{2}m^{3/2}$ by $\frac{3j}{4}mn^{3/4} - \frac{j}{4}n^{9/4}$ changes $S_{\mu,\nu}$ by at most $2\pi \frac{j}{4} \sum_{n \sim P} n^{-3/4} \ll jP^{1/4} \leq P^{7/24}$.

Step 3 (van der Corput A-process). With $H = P^{1/12}$, $|S_{\mu,\nu}|^2 \leq \frac{2P}{H} \sum_{0 \leq h < H} |T_h|$, where T_h sums $e(\Phi(n+2h) - \Phi(n))$ over the overlap and Φ is the linearized phase. The exact rewrite $m(n+2h)(n+2h)^{3/4} - m(n)n^{3/4} = g(n)(n+2h)^{3/4} + m(n)[(n+2h)^{3/4} - n^{3/4}]$ with $m(n) = n^{3/2} - \theta_n$ gives

$$\Phi(n+2h) - \Phi(n) = A_h(n) + \frac{3j}{4}g(n)(n+2h)^{3/4} + O(jhP^{-1/4}),$$

where A_h is smooth and the error is the $-\frac{3j}{4}\theta_n[(n+2h)^{3/4} - n^{3/4}]$ term. The two $n^{5/4}$ -scale contributions of A_h cancel at leading order, leaving $A_h'' = \frac{81}{256}jh^2n^{-7/4}(1 + o(1))$ plus an i -part $O(ihP^{-3/2})$, both $o(jhP^{-3/4})$ in these ranges.

Step 4 (cells). Split T_h over the $O(hP^{1/2})$ cells of Lemma 4.3(ii). On a cell, $g = G + \kappa$ with G constant and $\kappa(n) = [\{n^{3/2}\} \geq \rho(n)]$, $\rho = 1 - \{\delta(n)\}$ smooth and monotone there. Vaaler-expand κ with modes $e(rn^{3/2})$, $0 < |r| \leq R$. The moving endpoint costs nothing: since $\rho(n) = 1 + G - \delta(n)$, the coefficient's n -dependent piece contributes the exact smooth phase $r\delta(n)$, and $e(rn^{3/2} + r\delta(n)) = e(r(n+2h)^{3/2})$ up to a unimodular constant. Every r -term therefore falls into one of the two smooth families $e(rn^{3/2})$, $e(r(n+2h)^{3/2})$; the main term $\delta(n) - G$ has total variation exactly 1 per cell and is absorbed by one Abel summation.

Step 5 (second-derivative test per cell). The remaining phases are smooth. For the $r = 0$ parts, $|f''| \asymp \lambda_0 = jhP^{-3/4}$ with a single sign and bounded ratio across a cell; Lemma 3.3 on a cell of length $\ell \asymp P^{1/2}/h$ gives $\ll \ell\lambda_0^{1/2} + \lambda_0^{-1/2}$, hence over all cells $\ll (jh)^{1/2}P^{5/8} + j^{-1/2}h^{1/2}P^{7/8}$. For $r \neq 0$, the curvature is dominated by the r -term $\asymp |r|P^{-1/2}$ of fixed sign (as $jh \leq P^{1/8} \ll P^{1/4}$); the same test and the $1/|r|$ weights give $\ll R^{1/2}P^{3/4} + hP^{3/4} \log P$, and the majorant errors contribute $O(P/R) = O(P^{3/4})$. In total $|T_h| \ll P^{7/8}(1 + h^{1/2})$, uniformly in the modes.

Step 6 (assembly). $|S_{\mu,\nu}|^2 \ll P^2/H + P^{15/8}H^{1/2} \ll P^{23/12}$ at $H = P^{1/12}$, so $|S_{\mu,\nu}| \ll P^{23/24}$. Summing mode weights, majorants, and dyadic blocks gives the theorem with $N^{23/24} \log^3 N$. $\square$

The exponent $23/24$ is deliberately unoptimized; every stage has slack.

Proposition 4.5 (OE-branch third letter). For $a \in \{0, 1\}$,

$$\left| \sum_{n \leq N, n \text{ odd}} ((-1)^m)^a \psi(m^{1/2}) \right| \ll_\varepsilon N^{7/8+\varepsilon},$$

and consequently $\#\text{OEO}(N)$, $\#\text{OEE}(N) = N/8 + O(N^{7/8+\varepsilon})$. Together with Theorem 4.4 this makes depth 3 complete: each of OOO , OOE , OEO , OEE has density $1/8$ with a power saving.

Proof. Taylor expansion of $(X - \theta)^{1/2}$ with both correction terms one-signed gives

$$m^{1/2} = n^{3/4} + D_1(n), \quad -\frac{1}{2}X^{-1/2} - \frac{1}{8}(X-1)^{-3/2} \leq D_1 \leq 0.$$

D_1 is decaying, so replacing $\frac{l}{2}m^{1/2}$ by $\frac{l}{2}n^{3/4}$ inside a Vaaler mode costs $\ll l \sum_{n \sim P} n^{-3/4} \ll lP^{1/4}$, absorbable at $l \leq 2P^{1/24}$. The mode sums are then pure: $\sum_{n \sim P} e(\frac{i}{2}n^{3/2} + \frac{l}{2}n^{3/4})$ with $l \neq 0$. For $i \neq 0$, $\varphi'' \asymp |i|P^{-1/2}$ is single-signed and Lemma 3.3 gives $\ll i^{1/2}P^{3/4} + i^{-1/2}P^{1/4}$; for $i = 0$, $\varphi'' \asymp lP^{-5/4}$ gives $\ll l^{1/2}P^{3/8} + l^{-1/2}P^{5/8}$. Vaaler majorants, truncation tails, and dyadic assembly are as in Theorem 4.4, and every bound is $\ll P^{7/8}$ after summing mode weights. Branch consistency is Lemma 3.6: the $(1 + (-1)^m)/2$ factor vanishes exactly on odd m , where the true chain takes the $3/2$ -power branch. $\square$

Lemma 4.6 (fourth-letter linearization). For every odd $n \geq 3$,

$$v^{1/2} = n^{9/8} + D(n), \quad -\frac{3}{4}n^{-3/8} - n^{-9/8} \leq D(n) \leq 0.$$

Proof. Two applications of the Lemma 4.3(i) pattern. With $f(t) = (Y - t)^{1/2}$ on $[0, \theta_2]$: $v^{1/2} = m^{3/4} - \frac{1}{2}\theta_2 m^{-3/4} - E_2$, $0 \leq E_2 \leq \frac{1}{8}(Y-1)^{-3/2}$. With $f(t) = (X - t)^{3/4}$ on $[0, \theta]$: $m^{3/4} = n^{9/8} - \frac{3}{4}\theta n^{-3/8} - E_3$, $0 \leq E_3 \leq \frac{3}{32}(X-1)^{-5/4}$. Every term is nonpositive after the leading one, and $\frac{1}{2}m^{-3/4} + E_3 + E_2 \leq n^{-9/8}$ for $n \geq 3$. $\square$

Unlike the depth-2 layer, *every* non-smooth term now has decaying amplitude: the cumulative phase cost of replacing $\frac{k}{2}v^{1/2}$ by the smooth $\frac{k}{2}n^{9/8}$ over a dyadic block is $\ll kP^{5/8}$, negligible against the target.

Theorem 4.7 (triple parity discrepancy). For every $\varepsilon > 0$ and every $(a, b, c) \in \{0, 1\}^3 \setminus \{(0, 0, 0)\}$,

$$\left| \sum_{n \leq N, n \text{ odd}} \psi(n^{3/2})^a \psi(m^{3/2})^b \psi(v^{1/2})^c \right| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Consequently each of the eight sign classes of $(\psi(n^{3/2}), \psi(m^{3/2}), \psi(v^{1/2}))$ on odd $n \leq N$ has cardinality $\frac{N}{16} + O(N^{23/24+\varepsilon})$.

Proof. The $c = 0$ cases are Theorem 4.4 and Theorem 4.1. For $c = 1$, wave-expand all present sign factors as in Step 1 of Theorem 4.4, with truncations $J_1 = J_2 = J_3 = P^{1/24}$, and bound

$$S_{i,j,k}(P) = \sum_{n \in (P, 2P], n \text{ odd}} e(\frac{i}{2}n^{3/2} + \frac{j}{2}m^{3/2} + \frac{k}{2}v^{1/2})$$

uniformly for $|i| \leq 2J_1$, $|j| \leq 2J_2$, $1 \leq |k| \leq 2J_3$. By Lemma 4.6, replacing $\frac{k}{2}v^{1/2}$ by $\frac{k}{2}n^{9/8}$ costs $\ll |k|P^{5/8} \leq P^{2/3}$, before any differencing.

If $j = 0$, the remaining sum is a single smooth exponential sum with phase $\frac{i}{2}n^{3/2} + \frac{k}{2}n^{9/8}$. Both curvature terms are positive for positive modes, and for mixed signs with $|i| \geq 1$ the first dominates by $P^{3/8-1/24}$; Lemma 3.3 gives $\ll P^{5/8}$ for $i = 0$ and $\ll |i|^{1/2}P^{3/4} + P^{1/4}$ otherwise, both $\ll P^{23/24}$.

If $j \neq 0$, the phase is exactly the Theorem 4.4 phase plus the smooth passenger $\frac{k}{2}n^{9/8}$. Steps 2–6 run verbatim; the passenger modifies the smooth part of the differenced phase by $\frac{k}{2}[(n+2h)^{9/8} - n^{9/8}]$, whose second derivative $\ll |k|hP^{-15/8}$ is smaller than the retained cell curvature $jhP^{-3/4}$ and the r -mode curvature $|r|P^{-1/2}$ by powers of P . Every sign-dominance check of Theorem 4.4 holds with these margins. $\square$

Theorem 4.8 (the OE splits).**

$$\#OEOE(N), \#OEOO(N) = \frac{N}{16} + O(N^{7/8+\varepsilon}), \quad \#OEEE(N), \#OEEO(N) = \frac{N}{16} + O(N^{13/16+\varepsilon}).$$

Together with Theorems 4.4 and 4.7 this proves every depth-4 itinerary word class except OOO^* .

Proof. On the OE branch m is even and $w = \lfloor U \rfloor$, $U = m^{1/2}$. The w -level linearization is Lemma 4.3(i) verbatim with base U : since $Um^{1/4} = m^{3/4}$ exactly,

$$w^{3/2} = m^{3/4} - \frac{3}{2} m^{1/4} \theta_w + E, \quad 0 \leq E \leq \frac{3}{8} (U - 1)^{-1/2},$$

and $m^{3/4} = n^{9/8}$ up to one-signed corrections of scale $n^{-3/8}$. For a Vaaler mode $k \neq 0$, the phase is therefore $\varphi_1(n) - B(n)\theta_w(n)$ up to absorbable errors, with $\varphi_1 = \frac{i}{2}n^{3/2} + \frac{j}{2}n^{3/4} + \frac{k}{2}n^{9/8}$ and $B = \frac{3k}{4}n^{3/8}(1 + O(n^{-3/2}))$: a *single growing sawtooth* of amplitude $\asymp kn^{3/8}$ riding $\theta_w = \{m^{1/2}\}$, whose underlying integer increments once every $\asymp \frac{4}{3}n^{1/4}$ steps. Truncations: $J_1 = J_2 = J_3 = P^{1/8}$.

Drift-1 intervals. $B' \asymp kn^{-5/8}$, so B drifts by at most 1 on intervals I of length $L_0 = P^{5/8}/k$; there are $\asymp kP^{3/8}$ of them. On I , expand $e(-B\{U\}) = \sum_r a_r(B)e(rU)$ (Fourier in U , Vaaler truncation $|r+B| \leq T = P^{5/16}$); the coefficients satisfy $|a_r(B)| \leq \min(1, |r+B|^{-1})$ with window mass $O(\log T)$, and their n -dependence through $B(n)$ has total variation $O(1)$ per interval, removed by one partial summation per mode.

Mode sums. Smoothing $rU \rightarrow rn^{3/4}$ costs $kP^{5/8}$ in total. Writing $r = -B_0 + t$, $|t| \leq T$, the curvature of the mode phase is

$$\varphi'' = \frac{27k}{128}n^{-7/8} - \frac{3}{16}(t + \frac{j}{2})n^{-5/4} + \frac{3i}{8}n^{-1/2}.$$

For $i \neq 0$ the last term dominates and Lemma 3.3 gives $\ll i^{1/2}P^{3/4} + i^{-1/2}kP^{5/8}$ after summing intervals. For $i = 0$, $(T + J_2)P^{-5/4} \ll kP^{-7/8}$ because $T = P^{5/16} \ll kP^{3/8}$ for every $k \geq 1$, so $\lambda \asymp kn^{-7/8}$ is single-signed; Lemma 3.3 per interval and summation over $\asymp kP^{3/8}$ intervals give $S_k \ll k^{1/2}P^{13/16+\varepsilon}$. With Vaaler weights $1/k$, $\sum_{k \leq J_3} k^{-1/2}P^{13/16} = J_3^{1/2}P^{13/16} = P^{7/8}$, balanced against the truncation error $P/J_3 = P^{7/8}$ at $J_3 = P^{1/8}$. Total $\ll P^{7/8+\varepsilon}$.

OEE branch. The fourth letter needs $\psi(w^{1/2})$, and $w^{1/2} = U^{1/2} - \frac{1}{2}\theta_w U^{-1/2} - \frac{1}{8}\theta_w^2(U - \xi)^{-3/2}$ has *decaying* amplitudes only, with $U^{1/2} = m^{1/4} \rightarrow n^{3/8}$ likewise. The pure modes $e(\frac{l}{2}n^{3/8})$ have $\lambda \asymp lP^{-13/8}$, giving $l^{1/2}P^{3/16} + l^{-1/2}P^{13/16}$; the mixed cases sit in the dominance hierarchy $iP^{-1/2} \gg jP^{-5/4} \gg lP^{-13/8}$ with all three second derivatives of the same sign. Total $\ll N^{13/16+\varepsilon}$. $\square$

Corollary 4.9 (depth-4 census below the kernel; density 13/16). Every itinerary word class of depth at most four except OOO^* satisfies $\#\{n \leq N : \text{word}(n) \text{ has prefix } w\} = 2^{-|w|}N + O(N^{1-\delta_w})$ with the explicit exponents above. In particular

$$\#\{n \leq N : n \text{ odd, itinerary } OOOE\} = \frac{N}{16} + O(N^{23/24+\varepsilon}),$$

and the class of starts with a certified descent within four steps — evens (one step), OE (two steps), $OOEE$ (four steps) — has cardinality

$$\frac{N}{2} + \frac{N}{4} + \frac{N}{16} + O(N^{23/24+\varepsilon}) = \frac{13N}{16} + O(N^{23/24+\varepsilon}).$$

Proof. For $OOEE$, Lemma 3.6 gives $\#OOEE = \frac{1}{8} \sum_{n \leq N \text{ odd}} (1 - \psi_1)(1 + \psi_2)(1 + \psi_3)$ with $\psi_1 = \psi(n^{3/2})$, $\psi_2 = \psi(m^{3/2})$, $\psi_3 = \psi(v^{1/2})$; expanding gives the main term and seven sign sums bounded

by Theorem 4.7. Every *OOEE* start descends within four steps by Proposition 3.1: $3^2 < 2^4$. The even class and the *OE* class carry the one- and two-step certificates, and the three classes are disjoint. The remaining depth- ≤ 4 prefixes are Theorem 4.1 (depth 1), Theorem 4.4 and Proposition 4.5 (depths 2–3), and Theorems 4.7–4.8 (depth 4). $\square$

The density $13/16$ is the exact ceiling of this one-growing-layer machinery: a word contracts iff $3^o < 2^\ell$, the method so far proves letters at positions 1–3 of any word plus further letters along even branches only, and the contracting minimal words with all odd letters at positions ≤ 2 are exactly *E*, *OE*, and *OOEE*. Any further certified density requires the *OOO** split — a second growing layer, where the fourth-letter phase coefficient $W \asymp kn^{9/8}$ crosses integers within single steps and no drift-1 interval exists. Sections 5 and 6 close that split and harvest the contracting words it unlocks.

5. The kernel theorem

Every reorganization of the *OOO** phase funnels into one object: for smooth weights c of scale $kP^{9/8}$ on $n \sim P$,

$$K_c(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e(c(n) \{ \lfloor n^{3/2} \rfloor^{3/2} \}).$$

This section proves the kernel bound at full length: every estimate that the proof uses is displayed, with its dependence on the mode k , the differencing shifts h_1, h_2, h_3 , the branch offset j , and the expansion frequencies made explicit, and with numerical constants displayed at each of the sign-critical dominance checks. The expansion corrects one estimate of the earlier laboratory record and absorbs a second. First, the correction: the mixed pieces were previously modelled as $e(sX)$ with a frozen real coefficient $s \asymp qP^{3/4}$; in exact form they are level-2 waves $e(qY)$ times an optional frozen-floor factor, and the frozen model silently discards the sawtooth of amplitude $\asymp qP^{3/4}$ hidden in $Y = (X - \theta)^{3/2}$. Treated exactly — by the same exact linearization that proves Theorem 4.4 — the targeted third differencing survives, but the honest bound is $q^{-1/6} P^{23/24+\varepsilon}$ (Lemma 5.2), not $q^{1/6} P^{7/8+\varepsilon}$: the mixed piece is a depth-2 object and receives exactly depth-2 strength. The kernel saving therefore drops from the formerly claimed $\delta = \frac{1}{72}$ to $\delta = \frac{1}{96}$; every downstream exponent $1 - \frac{1}{72}$ becomes $1 - \frac{1}{96}$, and no density changes. Second, the absorbed observation: the pure smooth pieces of Step 4 carry a factor $(k|j|)^{1/2}$ that the earlier record did not display; since $(k|j|)^{1/2} P^{15/16} \leq P^{23/24}$ exactly on the standing range $k \leq P^{1/24}$, this loss sits below the new mixed-piece bottleneck and the corrected bound remains uniform in k on the original range.

The next lemma collects the exact reductions behind the treatment: the kernel phase is the level-2 local floor defect; the second difference of the level-2 integer obeys an exact floor identity; on an explicit carry-branch decomposition that second difference is a smooth function with a frozen floor; and the doubly differenced kernel phase decomposes exactly into four bounded pieces — the master identity — with no growing smooth part left to estimate.

Lemma 5.1 (level-2 defect, double gap, branch freeze, and the master identity). Fix shifts $d_1 = 2h_1$, $d_2 = 2h_2$ and write Δ_1 , Δ_2 , $\Delta\Delta$ for the corresponding difference operators in n , and $W = \Delta_1 Y$, $W' = \Delta_2 Y$.

(i) For odd $n \geq 3$,

$$\frac{3}{4} v^{1/2} \theta_2 = \frac{1}{2} (m^{9/4} - v^{3/2}) - R, \quad 0 \leq R \leq \frac{3}{16} v^{-1/2}.$$

Hence for $c = \frac{3k}{4} v^{1/2}$ the kernel phase $c\theta_2$ equals $\frac{k}{2} (Y^{3/2} - v^{3/2})$ up to $kR \ll kP^{-9/8}$ per term: the kernel is the exponential sum of the level-2 local floor defect.

(ii) Exactly, with $g_2 = \Delta_1 v$,

$$\Delta_2 g_2 = \lfloor \Delta \Delta Y \rfloor + \kappa'' + \Delta_2 \kappa_2, \quad \kappa_2 = [\theta_2 \geq 1 - \{W\}], \quad \kappa'' = [\{W\} \geq 1 - \{\Delta \Delta Y\}],$$

and every carry is a difference of unit sawtooths: $[\{A\} + \{B\} \geq 1] = \{A\} + \{B\} - \{A+B\}$ (Lean: `seq_floor_gap_second`).

(iii) Write $b_1 = \lfloor \Delta_1 X \rfloor$, $b_2 = \lfloor \Delta_2 X \rfloor$, $b_{12} = \lfloor \Delta_{12} X \rfloor$ (shift $d_1 + d_2$) — each constant on runs of length $\asymp P^{1/2}/h$ — and let $\kappa = (\kappa_1, \kappa_2, \kappa_{12}) \in \{0, 1\}^3$ be the level-1 carries, so that the shifted values of m are $m + \beta_1$, $m + \beta_2$, $m + \beta_{12}$ with $\beta_i = b_i + \kappa_i$. On each b -run intersection and carry branch,

$$\Delta \Delta Y = F_\kappa(m), \quad F_\kappa(m) = (m + \beta_{12})^{3/2} - (m + \beta_1)^{3/2} - (m + \beta_2)^{3/2} + m^{3/2}$$

exactly. The net offset $j = \beta_{12} - \beta_1 - \beta_2$ satisfies $|j| \leq 3$ for $h_1 h_2 \leq P^{1/2}/3$, and F splits exactly into an offset term and a genuine second difference: for some ξ_1 between 0 and j and some $\xi_2 \in (0, \beta_1 + \beta_2)$,

$$F_\kappa(m) = \frac{3}{2} j (m + \beta_1 + \beta_2 + \xi_1)^{1/2} + \frac{3}{4} \beta_1 \beta_2 (m + \xi_2)^{-1/2}.$$

Consequently, with $G := F_\kappa \circ X$,

$$\frac{3}{2} |j| P^{3/4} \leq \left| \frac{3}{2} j(\cdot)^{1/2} \right| \leq 2.6 |j| P^{3/4}, \quad 1.4 h_1 h_2 P^{1/4} \leq \frac{3}{4} \beta_1 \beta_2 (\cdot)^{-1/2} \leq 15 h_1 h_2 P^{1/4},$$

$$|G'(n)| \leq 2 |j| P^{-1/4} + 20 h_1 h_2 P^{-3/4} < 1, \quad |G''(n)| \leq 2 |j| P^{-5/4} + 25 h_1 h_2 P^{-7/4},$$

so $\lfloor G(n) \rfloor$ is constant on runs of length $\geq \frac{1}{22} \min(P^{1/4}/(|j|+1), P^{3/4}/(h_1 h_2))$. The branch indicator is a finite union of arcs in the single variable θ with slowly moving endpoints (drifts $\leq 1.5 h P^{-1/2}$ per step): exactly the moving-endpoint pattern of Step 4 of Theorem 4.4, absorbed by the same exact shift device at $O(\log P)$ mode-mass cost.

(iv) (*Master identity.*) With $\kappa_2 = [\theta_2 \geq 1 - \{W\}]$, $\kappa'_2 = [\theta_2 \geq 1 - \{W'\}]$ the level-2 carries at the two shifts, and κ'' , $\Delta_2 \kappa_2$ as in (ii), the doubly differenced kernel phase decomposes exactly:

$$\Delta \Delta(c \theta_2) = (\Delta \Delta c) \theta_2 + (\Delta_2 c)(n + d_1) (\{W\} - \kappa_2) + (\Delta_1 c)(n + d_2) (\{W'\} - \kappa'_2) + c_{11} (\{\Delta \Delta Y\} - \kappa'' - \Delta_2 \kappa_2),$$

where c_{11} is the doubly shifted weight. Every bracket is bounded by 2 in absolute value: no unbounded smooth part survives the differencing.

Proof. (i) Taylor of $(v + \theta_2)^{3/2}$ at v , with $Y^{3/2} = m^{9/4}$. (ii) The gap identity of Lemma 4.3(ii) applied twice — to Y at shift d_1 , then to the real sequence $n \mapsto W(n)$ at shift d_2 , with $\Delta_2 W = \Delta \Delta Y$; the sawtooth form of the carry is the Lean-verified floor-carry identity rearranged. (iii) The corner identities are Lemma 4.3(ii) at the three shifts, and the offset bound follows from $|\Delta \Delta X| \leq 4 h_1 h_2 \sup |X''| = 3 h_1 h_2 P^{-1/2} < 1$. For the split, group $F = [(m + \beta_1 + \beta_2 + j)^{3/2} - (m + \beta_1 + \beta_2)^{3/2}] + [(m + \beta_1 + \beta_2)^{3/2} - (m + \beta_1)^{3/2} - (m + \beta_2)^{3/2} + m^{3/2}]$: the first bracket is $\frac{3}{2} j(\cdot)^{1/2}$ by the mean value theorem, the second is $\beta_1 \beta_2 f''(m + \xi_2)$ for $f = x^{3/2}$ by the double mean value theorem. The displayed numerical bounds use $m \in (P^{3/2} - 1, 2^{3/2} P^{3/2}]$, $\beta_i \in [3 h_i P^{1/2} - 1, 3 \sqrt{2} h_i P^{1/2} + 1]$ (from $\Delta_i X = 2 h_i X'(\xi)$ and (E1) below), and $G' = F'(X) X'$, $G'' = F''(X) X'^2 + F'(X) X''$ with $F' = \frac{3}{4} j(\cdot)^{-1/2} - \frac{3}{8} \beta_1 \beta_2 (\cdot)^{-3/2}$, $F'' = -\frac{3}{8} j(\cdot)^{-3/2} + \frac{9}{16} \beta_1 \beta_2 (\cdot)^{-5/2}$. (iv) By (ii) and the definitions: $\Delta_1 \theta_2 = \Delta_1 Y - \Delta_1 v = W - (\lfloor W \rfloor + \kappa_2) = \{W\} - \kappa_2$ (Lemma 4.3(ii) applied to Y), likewise $\Delta_2 \theta_2 = \{W'\} - \kappa'_2$, and $\Delta \Delta \theta_2 = \Delta \Delta Y - \Delta_2 g_2 = \{\Delta \Delta Y\} - \kappa'' - \Delta_2 \kappa_2$ by (ii). Substitute these into the exact product rule

$$\Delta \Delta(c f) = c_{11} \Delta \Delta f + (\Delta_2 c)(n + d_1) \Delta_1 f + (\Delta_1 c)(n + d_2) \Delta_2 f + (\Delta \Delta c) f.$$

The identity is machine-validated on 12,000 exact scaled-integer samples across shift pairs $(h_1, h_2) \in \{(1, 1), (2, 3), (5, 7)\}$ (`master_identity_check`), on top of the Lean-anchored constituents (`seq_floor_gap`, `seq_floor_gap_second`). $\square$

One warning, essential to the organization: $\lfloor \Delta \Delta Y \rfloor$ itself is *not* frozen. The level-1 carry toggles at essentially every step and shifts $\Delta \Delta Y$ by $\frac{3}{2}j_1 m^{1/2}$, a jump of size $\asymp P^{3/4}$. The branch decomposition of (iii), which carries the flicker inside the indicator over the eight carry branches, is therefore forced: no decomposition conditioning on the *values* of the level-1 gaps produces sets on which the second difference is smooth.

Standing estimates

All estimates of this section take place on a dyadic block $n \in (P, 2P]$, n odd, under the standing constraints

$$(C1) \quad kh_1 h_2 \leq P^{1/8}, \quad (C2) \quad h_1 h_2 \leq P^{1/2}/3, \quad (C3) \quad 1 \leq k \leq P^{1/24},$$

and every displayed constant below is valid for $P \geq P_0$ with an absolute P_0 . The base derivatives are

$$(E1) \quad X' = \frac{3}{2}n^{1/2} \in (1.5, 2.13] P^{1/2}, \quad X'' = \frac{3}{4}n^{-1/2} \in [0.53, 0.75] P^{-1/2}, \\ -X''' = \frac{3}{8}n^{-3/2} \in [0.13, 0.375] P^{-3/2}, \quad X'''' = \frac{9}{16}n^{-5/2} \leq 0.57 P^{-5/2},$$

and $m = X - \theta \in (P^{3/2} - 1, 2.83 P^{3/2}]$. Every gap is a mean value: for any shift $2h$,

$$(E2) \quad \Delta X = 2hX'(\xi) \in (3hP^{1/2}, 4.3hP^{1/2}), \quad \beta := \lfloor \Delta X \rfloor + \kappa \in (3hP^{1/2} - 1, 4.3hP^{1/2} + 1),$$

so that every level-1 gap integer at shift $2h$ is $\asymp hP^{1/2}$ with the displayed constants. For the monomial weight $c = \frac{3k}{4}n^{9/8}$ (the general c of Theorem 5.3 changes only the absolute constants),

$$(E3) \quad c' = \frac{27}{32}kn^{1/8}, \quad c'' = \frac{27}{256}kn^{-7/8}, \quad c''' = -\frac{189}{2048}kn^{-15/8}, \quad c'''' = \frac{2835}{16384}kn^{-23/8},$$

$$(E4) \quad \Delta_i c = 2h_i c'(\xi) \in (1.68, 1.85) kh_i P^{1/8}, \quad \Delta \Delta c = 4h_1 h_2 c''(\xi) \in (0.22, 0.43) kh_1 h_2 P^{-7/8}.$$

On the level-2 side, with β_i as in Lemma 5.1(iii),

$$(E5) \quad W = \Delta_1 Y \in (4.4, 11) h_1 P^{5/4}, \quad |(W \circ \text{cell})'| = |W'_1(m) X'| \in (2.2, 10.4) h_1 P^{1/4},$$

where $W_1(m) = (m + \beta_1)^{3/2} - m^{3/2}$, so $\{W\}$ is a *fast* sawtooth (its argument moves by $\gg 1$ per step), while θ , $\{\Delta_i X\}$, and $\{\Delta \Delta Y\}$ are slow. The branch bounds of Lemma 5.1(iii) give, per branch with offset j ,

$$(E6) \quad |(cF)''| = \frac{945}{512} k|j| n^{-1/8} (1 + O(P^{-1/4})) + O(kh_1 h_2 P^{-5/8}),$$

the two shapes being the composite monomials $kjn^{15/8}$ (offset branches) and $kh_1 h_2$ -scale $n^{11/8}$ (zero-offset), with sign product $\alpha(\alpha - 1)(\alpha + \beta - 2)(\alpha + \beta - 3) > 0$ at $\alpha = \frac{9}{8}$, $\beta \in \{\frac{3}{4}, \frac{1}{4}\}$. Finally, the run and cell inventories: level-1 gap cells at shift $2h$ number at most $1.5hP^{1/2} + 1$ with lengths in $[\frac{2}{3}, 0.95] P^{1/2}/h$, and the frozen-floor runs of $\lfloor F_\kappa(X) \rfloor$ number at most $22(|j| + 1)P^{3/4}$ by the derivative bound of Lemma 5.1(iii).

Throughout, “ A is dominated by B at margin $P^{-\rho}$ ” means $\sup |A|/\inf |B| \leq CP^{-\rho}$ with the displayed C : the composite second derivative then keeps the sign and, up to the factor $1 + CP^{-\rho}$, the size of B , so Lemma 3.3 applies at B ’s scale.

The proof of the kernel theorem classifies the differenced pieces into four classes; three are handled by standard tests, and the fourth — an exact level-2 wave $e(qY)$, possibly riding a frozen floor — is the genuinely new difficulty. We isolate it as a standalone lemma with its own differencing, so that it can be checked independently.

Lemma 5.2 (level-2 waves: the mixed-piece bound). Assume (C1)–(C3), write $\mathcal{D} = \{0, d_1, d_2, d_1+d_2\}$, and call a *decoration* any sum ρ of at most four terms of the classes

- (D1) $q' \Delta_{2h} \Delta_{2h'} Y(n+d')$ with $|q'| \leq P^{1/16}$, $1 \leq h' \leq P^{1/24}$, $d' \in \mathcal{D}$;
- (D2) $-\Delta_{2h}(c[F_\kappa(X)])(n)$, with F_κ a branch function of Lemma 5.1(iii), offset $|j| \leq 3$;
- (D3) a smooth φ with $|\varphi''| \leq 3kh_1h_2P^{-5/8}$ and $|\varphi'''| \leq 3kh_1h_2P^{-13/8}$;

here $2h$ is the shift of part (i), and in part (ii) the decorations are read after the differencing there. Then, for sums over $n \in (P, 2P]$, n odd:

- (i) (*differenced wave*) for all integers $u, h \geq 1$ with $h \leq P^{1/8}$, $uh \leq P^{1/2}$, and $d \in \mathcal{D}$,

$$V := \left| \sum_n e(u \Delta_{2h} Y(n+d) + \rho(n)) \right| \ll \left((uh)^{1/2} P^{5/8} + (h/u)^{1/2} P^{7/8} + P^{7/8} \right) P^\varepsilon;$$

- (ii) (*wave*) for all integers q, q' with $|q|, |q'| \leq P^{1/16}$ and $t := q+q' \neq 0$, all $d \neq d' \in \mathcal{D}$, $\varepsilon_0 \in \{0, 1\}$, and φ of class (D3),

$$U := \left| \sum_n e(qY(n+d) + q'Y(n+d') - \varepsilon_0 c(n)[F_\kappa(X(n))] + \varphi(n)) \right| \ll |t|^{-1/6} P^{23/24+\varepsilon}.$$

Part (ii) is the mixed-piece bound proper: the level-2 wave $e(qY)$, possibly riding a frozen floor. Part (i) is the engine that (ii)'s targeted differencing feeds, and is also used directly for the $\{W\}$ -content of the kernel proof. The exponent $\frac{23}{24}$ is not an accident: written exactly, the wave $e(qY)$ is the depth-2 object of Theorem 4.4, and part (ii) recovers exactly the depth-2 strength — no more. The earlier laboratory record modelled U as $e(sX)$ with s frozen of size $qP^{3/4}$ and claimed $q^{1/6}P^{7/8+\varepsilon}$; that model discards the sawtooth $-\frac{3}{2}qX^{1/2}\theta$ of amplitude $\asymp qP^{3/4}$ inside $qY = q(X-\theta)^{3/2}$, and the claim does not survive the exact treatment.

Proof of (ii) from (i). Replacing the summand by its conjugate we may take $t \geq 1$. Write, exactly,

$$qY(n+d) + q'Y(n+d') = tY(n+d) + q' \Delta_{|d'-d|} Y(n+\min(d, d')),$$

so the second term is a decoration seed. Weyl differencing at shift $2h_3$, $1 \leq h_3 \leq H_3 := \lceil t^{1/3} P^{1/12} \rceil$:

$$|U|^2 \leq \frac{P^2}{H_3} + \frac{2P}{H_3} \sum_{h_3=1}^{H_3} |V_{h_3}|, \quad V_{h_3} = \sum_n e(t \Delta_{2h_3} Y(n+d) + \rho_{h_3}(n)),$$

where ρ_{h_3} collects $q' \Delta_{2h_3} \Delta_{|d'-d|} Y$ (class (D1) with $h' = |d' - d|/2 \leq h_1+h_2 \leq P^{1/24} \cdot 2$), $-\varepsilon_0 \Delta_{2h_3}(c[F(X)])$ (class (D2)), and $\Delta_{2h_3} \varphi$ (class (D3): $|(\Delta_{2h_3} \varphi)''| \leq 2h_3 \sup |\varphi'''| \leq 6kh_1h_2h_3P^{-13/8} \leq 3kh_1h_2P^{-5/8}$ since $h_3 \leq P^{1/4}$). The hypothesis of (i) holds: $th_3 \leq t \cdot 2t^{1/3} P^{1/12} = 2t^{4/3} P^{1/12} \leq 2P^{1/12+1/12} \leq P^{1/2}$. Part (i) gives

$$|U|^2 \leq \frac{P^2}{H_3} + \left(4t^{1/2} H_3^{1/2} P^{13/8} + 4t^{-1/2} H_3^{1/2} P^{15/8} + 4P^{15/8} \right) P^\varepsilon,$$

and at $H_3 = \lceil t^{1/3} P^{1/12} \rceil$ the four terms are, in order, $\leq t^{-1/3} P^{23/12}$; $\leq 6t^{2/3} P^{5/3} = 6(tP^{-1/4})t^{-1/3} P^{23/12} \leq 6P^{1/16-1/4} \cdot t^{-1/3} P^{23/12}$; $\leq 6t^{-1/3} P^{1/24} P^{15/8} = 6t^{-1/3} P^{23/12}$; and $\leq 4(t^{1/3} P^{-1/24})t^{-1/3} P^{23/12} \leq 4P^{1/48-1/24} t^{-1/3} P^{23/12}$. Hence $|U|^2 \ll t^{-1/3} P^{23/12+\varepsilon}$, which is (ii).

Proof of (i). Six stages; write $\nu = n + d$ (a shift of the block by at most $d_1 + d_2 \leq P^{1/2}$, harmless in every estimate) and $\Delta = \Delta_{2h}$.

Stage 1 (the m -linear form). By Lemma 4.3(i) at the bases ν and $\nu+2h$,

$$u \Delta Y = u A_h(\nu) + \frac{3u}{2} g(\nu) (\nu+2h)^{3/4} - \frac{3u}{2} \theta_\nu \Delta(\nu^{3/4}) + u \Delta E,$$

with $g = m(\nu+2h) - m(\nu)$, $|u \Delta E| \leq u P^{-3/4}$ (total cost over the block $\leq 2\pi u P^{1/4} \leq 2\pi P^{3/4}$), and

$$A_h(\nu) = \frac{3}{2} \nu^{3/2} \Delta(\nu^{3/4}) - \frac{1}{2} \Delta(\nu^{9/4}) = -\frac{27}{8} h^2 \nu^{1/4} (1 + O(hP^{-1})) :$$

the two $\nu^{5/4}$ -scale contributions cancel exactly at leading order (both equal $\frac{9}{4} h \nu^{5/4}$), leaving $|u A_h''| \leq 0.64 u h^2 P^{-7/4}$. The sawtooth coefficient is

$$B(\nu) := \frac{3u}{2} \Delta(\nu^{3/4}) = \frac{9}{4} u h \xi^{-1/4} \in (1.89, 2.25] u h P^{-1/4}.$$

Stage 2 (gap cells and the exact shift device). Split into the level sets of $\lfloor \delta_h \rfloor$, $\delta_h(\nu) = (\nu+2h)^{3/2} - \nu^{3/2}$: at most $1.5hP^{1/2} + 1$ cells of length in $[\frac{2}{3}, 0.95] P^{1/2}/h$ (E1). On a cell $g = G + \kappa$ with $G = \lfloor \delta_h \rfloor$ frozen and $\kappa = [\theta_\nu \geq 1 - \{\delta_h\}]$; Vaaler-expand κ at truncation $R_0 = P^{1/4}$ (majorant cost $\leq 4P/R_0 = 4P^{3/4}$). The moving endpoint costs nothing: as in Theorem 4.4, Step 4, the endpoint's smooth part contributes the exact phase $r\delta_h(\nu)$, and $e(r\nu^{3/2} + r\delta_h(\nu)) = e(r(\nu+2h)^{3/2})$, so every r -term falls into the two smooth families $e(r\nu^{3/2})$, $e(r(\nu+2h)^{3/2})$, $0 < |r| \leq R_0$, with weights $\leq 1/|r|$; the in-cell variation of the coefficient is exactly 1 and one Abel summation absorbs it at a factor $\leq 2+2\pi$.

Stage 3 (the θ -sawtooth). Two regimes. (s1) If $uh \leq P^{3/16}$ then $|B| \leq 2.25P^{-1/16} < \frac{1}{2}$ for $P \geq P_0$: Lemma 3.8 with $T = P^{1/2}$, $J = R_0$ expands $e(-B_0\{\nu^{3/2}\} \dots)$ per window (here one window: the total drift of B is $\leq 0.6uhP^{-1/4} \leq 0.6P^{-1/16} < 1$) into the same two mode families at coefficient factor $\min(2, 2\pi|B|) \leq 14P^{-1/16}$, mass $O(\log P)$, flat cost $\leq 12P^{1/2}$, majorant $\leq 14P^{-1/16} \cdot 4P^{3/4}$. (s2) If $P^{3/16} < uh \leq P^{1/2}$ then $|B| \leq 2.25P^{1/4}$ and B drifts by at most 1 on windows of length $\geq P^{5/4}/(0.6uh) \geq 1.2P^{3/4}$: at most $0.6P^{1/4} + 1$ windows. Per window Lemma 3.8 at the centre B_0 ($T = P^{1/2}$: flat cost $\leq 8(1+2.25P^{1/4})P^{1/2} \leq 27P^{3/4}$ in total) produces modes $e(w\nu^{3/2})$ whose coefficients decay as $\min(2, \frac{1}{\pi|w+B_0|}) + \min(2, \frac{1}{\pi|w|})$; the window-boundary cost is, using $uh > P^{3/16}$, $\leq (0.6P^{1/4}+1) \cdot 1.83(0.30uh)^{-1/2} P^{3/8} \leq 2.1 P^{1/4-3/32+3/8} = 2.1 P^{17/32} \leq P^{5/8}$. Modes with $|w|$ in the collision window are treated in Stage 5.

Stage 4 (the main curvature, $r = w = 0$). On a cell the remaining phase is $\frac{3u}{2} G(\nu+2h)^{3/4} + u A_h +$ (smooth decorations), with second derivative

$$-\frac{9}{32} u G(\nu+2h)^{-5/4} \in -[0.30, 1.35] u h P^{-3/4} \quad (\text{E2}),$$

single-signed with in-cell ratio $\sup/\inf \leq 4.5$; this is the composite whose two contributions $-\frac{27}{32} \beta n^{-5/4}$ and $+\frac{9}{16} \beta n^{-5/4}$ (net $-\frac{9}{32} \beta n^{-5/4}$, margin 3:2) are machine-validated (`kernel_margin_scan`, gate m1). The competitor $u A_h''$ has ratio $\leq 0.64 u h^2 P^{-7/4} / (0.30 u h P^{-3/4}) \leq 2.2 h P^{-1} \leq 2.2 P^{-3/4}$; decoration competitors are bounded in Stage 6. Lemma 3.3 per cell and summation give

$$\sum_{\text{cells}} \left(\ell_i \lambda^{1/2} + \lambda^{-1/2} \right) \leq 2.3 (uh)^{1/2} P^{5/8} + 2.8 (h/u)^{1/2} P^{7/8}.$$

Stage 5 (nonzero modes and collisions). A mode $w \neq 0$ (or $r \neq 0$; identical treatment with weight $1/|r|$) adds the phase $w\nu^{3/2}$ of curvature $\geq 0.53|w|P^{-1/2}$. If $0.53|w|P^{-1/2} \geq 4 \cdot 1.35uhP^{-3/4}$, i.e. $|w| \geq 10.2uhP^{-1/4}$, the mode curvature dominates at margin ≥ 4 : Lemma 3.3 over the full block gives $\leq 1.4|w|^{1/2}P^{3/4} + 1.4|w|^{-1/2}P^{1/4}$, and the coefficient-weighted sums are $\leq 3R_0^{1/2}P^{3/4}\log P = 3P^{7/8}\log P$ (families of Stage 2, regime (s1)) and $\leq CP^{7/8}\log P$ (regime (s2) tails). If $|w| \leq 0.1uhP^{-1/4}$ the main curvature dominates at margin ≥ 4 and the mode rides along at Stage 4's scale, weight summable to $O(\log P)$. In the remaining collision window $0.1uhP^{-1/4} \leq |w| \leq 10.2uhP^{-1/4}$ (nonempty only in regime (s2)), the two curvatures can cancel: there the phase is the two-term monomial $av^{3/4} + w\nu^{3/2}$ with $a = \frac{3u}{2}G \asymp uhP^{1/2}$ plus perturbations already shown ρ_0 -small, and Lemma 3.9 with $(\alpha, \beta) = (\frac{3}{4}, \frac{3}{2})$ applies per window with $M \leq 1.35uhP^{-3/4} \leq 1.35P^{-1/4}$:

$$\leq |I_w|M^{1/2} + M^{-1/2} + |I_w|(M/P)^{1/6} + |I_w|^{1/2}(M/P)^{-1/6},$$

which, summed over windows with the $O(\log P)$ collision-window coefficient mass, is $\leq C((uh)^{1/2}P^{5/8} + (uh)^{1/6}P^{17/24} + (uh)^{-1/6}P^{19/24} + P^{5/8})\log P \leq CP^{7/8}\log P$ (using $uh \leq P^{1/2}$).

Stage 6 (decorations). Each class is dominated at a displayed margin against the Stage-4 curvature $\geq 0.30uhP^{-3/4}$.

- (D1). On the branch decomposition of Lemma 5.1(iii) at the shift pair (h_3 -role h, h'): the arcs are absorbed by the shift device (log mass); the b -run boundaries add $\leq 1.5(h+2h')P^{1/2} + 2$ cells, at boundary cost $\leq 1.5(h+2h')P^{1/2} \cdot 1.83(0.30uh)^{-1/2}P^{3/8} \leq 5.1(h/u)^{1/2}P^{7/8} + 10.3h'(uh)^{-1/2}P^{7/8}$ (the second term is what the H_3 -averaging of part (ii) absorbs: it contributes $\leq 2h't^{-2/3}P^{43/24} \leq P^{1/24-1/8}t^{-1/3}P^{23/12}$ there). The θ -coefficient of the decoration is $\leq |q'|(2|j'|P^{-1/4} + 20hh'P^{-3/4}) \leq 6P^{1/16-1/4} + 20P^{1/16+1/4+1/24-3/4} < 1$: sub-unit, expanded in the Stage-2 families. Its smooth curvature has ratio $\leq |q'|(2|j'|P^{-5/4} + 25hh'P^{-7/4})/(0.30uhP^{-3/4}) \leq 20P^{1/16}P^{-1/2} + 84P^{1/16+1/24}P^{-1} \leq P^{-1/4}$: dominated.
- (D2). Write $\Delta(c[F]) = (\Delta c)[F] + c_+\Delta[F]$ with $c_+ = c(\cdot+2h)$.
 - (a) $(\Delta c)[F] = (\Delta c)F - (\Delta c)\{F\}$: the smooth part has curvature $\leq ((\Delta c)F)'' \leq 7kh|j|P^{-9/8} + 28khh_1h_2P^{-13/8}$, ratio to the main curvature $\leq 24k|j|P^{-3/8}/u \leq 72P^{1/24-3/8}$: dominated. The sawtooth part has coefficient $\Delta c \in (1.68, 1.85)khP^{1/8}$, handled per windows of drift ≤ 1 : at most $2khP^{1/8}+1$ windows, at boundary cost $\leq 2khP^{1/8} \cdot 1.83(0.30uh)^{-1/2}P^{3/8} \leq 7k(h/u)^{1/2}P^{1/2}$. Per window, Lemma 3.8 at $T = P^{1/2}$ yields modes $e(q''(F \circ X))$ of curvature $\leq (2khP^{1/8}+P^{1/2}) \cdot 3|j|P^{-5/4} \leq 18P^{-3/4} \cdot P^{-1/16}$, ratio to the main curvature $\leq 60P^{-1/16}$: dominated. The flat cost is, per point, $8(1+B_0)/T$, so in total $\leq 8(1+1.85khP^{1/8})P^{1/2} \leq 8P^{1/2} + 15khP^{5/8} \leq 23P^{1/24+1/8+5/8} = 23P^{19/24} \leq P^{7/8}$, using $h \leq P^{1/8}$ and (C3).
 - (b) $c_+\Delta[F]$: by the gap identity applied to the sequence $F \circ X$, $\Delta[F] = [\Delta F] + \kappa_F$. ΔF has total drift $\leq P \cdot 2h \sup |(F \circ X)''| \leq P \cdot 2h(2|j|P^{-5/4} + 25h_1h_2P^{-7/4}) \leq 13hP^{-1/4} + 50hh_1h_2P^{-3/4} < 1$, so $[\Delta F]$ takes at most two values, on at most two intervals; per value the phase $-q_0c_+$ ($|q_0| \leq 2+6hP^{-1/4} \leq 3$) has curvature $\leq 0.4kP^{-7/8}$, ratio $\leq 1.4kP^{-1/8}/(uh) \leq 1.4P^{1/24-1/8}$: dominated. The carry $\kappa_F = \{F\} + \{\Delta F\} - \{F+\Delta F\}$ is a sum of unit sawtooths in slow variables (all drifts < 1 per step): Lemma 3.5 at truncation $P^{1/8}$ (majorant $\leq 3 \cdot 4P^{7/8}$) gives modes $e(q''(F \circ X))$ -type of curvature $\leq P^{1/8} \cdot 3|j|P^{-5/4} \leq 9P^{-9/8}$, ratio $\leq 30P^{-3/8}$: dominated.
- (D3). Ratio $\leq 2h \sup |\varphi'''|/(0.30uhP^{-3/4}) \leq 20kh_1h_2P^{-7/8}/u \leq 20P^{1/8-7/8}$: dominated.

Totals. Stages 1–6 bound V by

$$C\left((uh)^{1/2}P^{5/8} + (h/u)^{1/2}P^{7/8} + P^{7/8} + k(h/u)^{1/2}P^{1/2}\right)\log^3 P,$$

and the fourth term is $\leq P^{1/24} \cdot (h/u)^{1/2} P^{1/2}$, absorbed by the second. This is (i) up to the stated P^ε . $\square$

The balance to keep in mind: with the trivial bound $|V_{h_3}| \leq P$, part (ii)'s differencing returns nothing beyond the choice of H_3 ; the content of the lemma is that the floor, coefficient, and carry bookkeeping of Stages 3–6 survives the targeted differencing with the Stage-4 curvature intact. What is *not* available is the frozen-coefficient shortcut: per frozen run the run length $P^{1/4}$ is shorter than $\lambda^{-1/2} \asymp (uh)^{-1/2} P^{3/8}$ for small uh , and the run-boundary cost $(h/u)^{1/2} P^{7/8}$, summed against the H_3 -average, is exactly what pins the final exponent at $\frac{23}{24}$.

Theorem 5.3 (kernel cancellation). Let c be smooth on $(P, 2P]$ with $c^{(r)} \asymp kP^{9/8-r}$ for $r = 0, \dots, 4$, derivative signs following the monomial pattern (e.g. $c = \frac{3k}{4}n^{9/8}$), and $1 \leq k \leq P^{1/24}$. Then

$$K_c(P) \ll P^{1-1/96+\varepsilon},$$

uniformly in k .

Proof. Work on a dyadic block $n \sim P$, odd.

Step 1 (double Weyl differencing). Set $H_1 = P^{1/48}$, $H_2 = P^{1/24}$. The classical inequality $|\sum_n a_n|^2 \leq \frac{P+2H}{H} \sum_{|h|<H} (1 - \frac{|h|}{H}) \sum_n a_{n+2h} \overline{a_n}$ applied twice gives

$$|K_c|^2 \leq \frac{2P^2}{H_1} + \frac{4P}{H_1} \sum_{h_1=1}^{H_1} |T_1(h_1)|, \quad |T_1|^2 \leq \frac{2P^2}{H_2} + \frac{4P}{H_2} \sum_{h_2=1}^{H_2} |T_2(h_1, h_2)|,$$

with $T_2 = \sum_n e(\varphi_2)$, $\varphi_2 = \Delta\Delta(c\theta_2)$. For all $h_1 \leq H_1$, $h_2 \leq H_2$, $k \leq P^{1/24}$: $kh_1h_2 \leq P^{1/24+1/48+1/24} = P^{5/48} \leq P^{1/8}$, so (C1)–(C3) hold with room $P^{-1/48}$. We prove

$$|T_2| \ll P^{23/24+\varepsilon} \quad \text{uniformly in } h_1, h_2, k;$$

then $|T_1| \ll P^{1-1/48+\varepsilon}$ and $|K_c| \ll P^{1-1/96+\varepsilon}$, the three savings balancing exactly at the chosen H_1, H_2 .

Step 2 (master identity; the trivial piece). By Lemma 5.1(iv), exactly,

$$\varphi_2 = \underbrace{(\Delta\Delta c)\theta_2}_{M_1} + \underbrace{(\Delta_2 c)(n+d_1)(\{W\} - \kappa_2)}_{M_2} + \underbrace{(\Delta_1 c)(n+d_2)(\{W'\} - \kappa'_2)}_{M_3} + \underbrace{c_{11}(\{\Delta\Delta Y\} - \kappa'' - \Delta_2\kappa_2)}_{M_4}.$$

By (E4), $|M_1| \leq 0.43kh_1h_2P^{-7/8}$, so $|e(M_1) - 1| \leq 2.7kh_1h_2P^{-7/8}$ and deleting M_1 changes T_2 by at most $2.7kh_1h_2P^{1/8} \leq 2.7P^{1/4}$ by (C1). No growing smooth part remains: every other bracket is bounded by 2.

Step 3 (the expansion inventory). The three remaining pieces are expanded as follows; all truncations, masses, and error costs are displayed here once and summed in Step 6.

(3a) M_2 , *sawtooth part* $(\Delta_2 c)(n+d_1)\{W\}$. The coefficient $B(n) = \Delta_2 c(n+d_1) \in (1.68, 1.85)kh_2P^{1/8}$ drifts by $|B'| \leq 0.22kh_2P^{-7/8}$ per step: freeze it on at most $2kh_2P^{1/4}+1$ windows on which it moves by $\leq P^{-1/8}$ (residual cost $\sum_n |e((B-B_0)\{W\}) - 1| \leq 6.3P^{-1/8} \cdot P = 6.3P^{7/8}$). Per window, Lemma 3.8 at the centre B_0 with $T = P^{1/2}/(2h_1)$, $J = P^{1/4}$: flat cost in total $\leq 8(1+1.85kh_2P^{1/8}) \cdot 2h_1P^{1/2} \leq 16h_1P^{1/2} + 30kh_1h_2P^{5/8} \leq 46P^{3/4}$ by (C1); the majorant Δ_J -part costs $\leq 2 \cdot 4P/J = 8P^{3/4}$ plus J -mode sums at coefficients $\leq 1/(J+1)$; modes $e(uW)$ with mass $\leq C \log P$ and $u \leq$

$1.85kh_2P^{1/8} + P^{1/2}/(2h_1)$, so that $uh_1 \leq 1.85P^{1/4} + P^{1/2}/2 \leq P^{1/2}$: every mode is a Lemma 5.2(i) object at shift $h_1 \leq P^{1/48}$. Window boundaries cost $\leq 2kh_2P^{1/4} \cdot 3.4P^{3/8} \leq 7P^{17/24}$.

(3b) M_2 , *carry part* $(\Delta_2c)(n+d_1)\kappa_2$. Here $\kappa_2 \in \{0,1\}$ is an integer, so exactly $e(-(\Delta_2c)\kappa_2) = 1 + \kappa_2(e(-\Delta_2c) - 1)$: the large coefficient never multiplies a sawtooth inside an exponential. The factor $e(-\Delta_2c(n+d_1))$ is a smooth phase with $|(\Delta_2c)''| \leq 2h_2 \sup |c''| \leq 0.19kh_2P^{-15/8}$: class (D3). The weight $\kappa_2 = \{Y\} + \{W\} - \{Y+W\}$ (Lemma 5.1(ii)) is expanded *additively* as real numbers, each unit sawtooth by finite Fourier (Lemma 3.5) at truncation $J_2 = P^{1/24}$, coefficients $\leq 1/(\pi|q|)$, majorant cost $\leq 4P/J_2 = 4P^{23/24}$ per layer plus J_2 -mode averages. Since $Y+W = Y(n+d_1)$ exactly, the resulting modes are $e(qY(n))$, $e(qY(n+d_1))$ — Lemma 5.2(ii) objects — and $e(qW)$ — Lemma 5.2(i) objects.

(3c) M_3 . The mirror of (3a)–(3b) with $h_1 \leftrightarrow h_2$, producing $e(qY(n+d_2))$, $e(qW')$ -modes.

(3d) M_4 , *carries*. $\kappa'' = \{W\} + \{\Delta\Delta Y\} - \{W+\Delta\Delta Y\}$ with $W+\Delta\Delta Y = W(n+d_2)$ exactly, and $\Delta_2\kappa_2 = \kappa_2(n+d_2) - \kappa_2(n)$ with each κ_2 expanded as in (3b) at its base. The smooth factors $e(\mp c_{11})$ have $|c''| \leq 0.11kP^{-7/8}$: class (D3). New mode species: $e(q\{\Delta\Delta Y\}$ -content) — slow modes, treated in Step 5 — and $e(qY(n+d))$ for $d \in \{0, d_2, d_1+d_2\}$, $e(qW(n+d_2))$: Lemma 5.2 objects again.

(3e) M_4 , *the anchor*. The factor $e(c_{11}\{\Delta\Delta Y\})$ is never expanded away. On the branch decomposition of Lemma 5.1(iii) (arcs absorbed by the exact shift device at $O(\log P)$ mass and $4P^{3/4}$ majorant, as in Theorem 4.4, Step 4), $\{\Delta\Delta Y\} = G - J_F$ with $G = F_\kappa(X-\theta)$ and $J_F = \lfloor G \rfloor$ frozen on the runs of (E6). The anchor phase $c_{11}(G - J_F)$ is present in **every** final piece; its curvature scale ((E6), gate m2) is what orders the whole classification of Step 5.

After (3a)–(3e), every final piece of T_2 has the exact form

$$e\left(\underbrace{c_{11}(G - J_F)}_{\text{anchor}} + \underbrace{\sum_i q_i Y(n+e_i)}_{\text{waves, } e_i \in \mathcal{D}} + \underbrace{uW \text{ and } u'W' \text{-modes}}_{\text{differenced waves}} + \underbrace{\text{slow modes} + \varphi}_{(\text{D3})\text{-smooth}}\right)$$

with at most one mode from each expansion layer (three layers), and piece weights whose total mass is $O(\log^3 P)$ beyond the displayed majorant and flat costs.

Step 4 (pieces with wave content: Lemma 5.2(ii)). Let $t = \sum_i q_i$ be the total wave frequency, $|t| \leq 3J_2 \leq P^{1/16}$.

If $t \neq 0$: telescoping, $\sum_i q_i Y(n+e_i) = tY(n+e_1) + \sum_{i \geq 2} q_i \Delta_{e_i-e_1} Y(n+\min(e_1, e_i))$, and after the targeted differencing of Lemma 5.2(ii) the differenced-wave remainders and the u, u' -modes become (D1) decorations, the anchor becomes the (D2) decoration, and the smooth content is (D3): Lemma 5.2(ii) applies and gives $\ll |t|^{-1/6} P^{23/24+\varepsilon}$ per piece. The weight sum is

$$\sum_{t \neq 0} \sum_{q_1+q_2+q_3=t} \frac{1}{\pi^3 \max(1, |q_1|) \max(1, |q_2|) \max(1, |q_3|)} |t|^{-1/6} \ll \sum_{t \geq 1} \frac{\log^2(2+t)}{t^{7/6}} \ll 1,$$

so the total wave-piece contribution is $\ll P^{23/24+\varepsilon}$.

If $t = 0$: the wave content collapses exactly to differenced waves — e.g. $q(Y(n+d_2) - Y(n)) = qW'(n)$ — i.e. to (D1)-type resonant decorations of the remaining piece; these are handled in Step 5. This exactness is the reason no near-resonant analysis is ever needed for the wave frequencies: the resonant combination is an identity, not an approximation.

Step 5 (pieces without wave content: the anchor classes). The remaining pieces carry the anchor, at most two differenced-wave modes uW , $u'W'$ (including the resonant remnants of Step 4, which by

Lemma 5.1(iii) are smooth-per-branch with sub-unit θ -coefficients), slow modes, and (D3)-smooth content. Split by the anchor branch type.

(5a) Offset branches ($j \neq 0$). The anchor curvature is, by (E6) and the window-centre composite,

$$\lambda_a = \frac{729}{512} k|j| n^{-1/8} (1 + O(P^{-1/8})) \in [1.2, 1.5] k|j| P^{-1/8},$$

where $\frac{729}{512} = \frac{945}{512} - \frac{27}{64}$ is the sum of the smooth part $(cF)''$ and the window-centre mode uX'' at $u = -B(n_0)$: single-signed with ratio $\frac{945}{512} : \frac{27}{64} = 4.375$ (machine gate m2 of `kernel_margin_scan`). Every competitor is dominated at a displayed margin: differenced-wave modes $\leq 0.84 u h_1 P^{-3/4} \leq 0.51 P^{-1/4}$ (since $u h_1 \leq 0.6 P^{1/2}$), ratio $\leq 0.43 P^{-1/8}$ against $\lambda_a \geq 1.2 P^{-1/8}$; resonant (D1) content $\leq 6 |q'| P^{-5/4} \cdot P^{1/16}$ -scale, ratio $\leq 5 P^{-9/16}$; slow modes $\leq 3 J_2 |j| P^{-5/4}$, ratio $\leq 8 P^{1/24-9/8}$; (D3) content, ratio $\leq 3 h_1 h_2 P^{-1/2} \leq P^{-1/4}$. The θ -sawtooth of the anchor has coefficient $\frac{9}{16} k|j| P^{3/8}$ -scale with per-step drift $\leq 0.2 k|j| P^{-5/8} < 1$: at most $1.2 k|j| P^{3/8} + 1$ windows of length $\geq 0.8 P^{5/8} / (k|j|)$, window-boundary cost $\leq 1.2 k|j| P^{3/8} \cdot 0.92 (k|j|)^{-1/2} P^{1/16} \leq 1.1 (k|j|)^{1/2} P^{7/16}$. Off the collision band, window modes w are dominated at margin ≥ 4 either way and ride along or are estimated by Lemma 3.3 at their own scale with $1/|w+B_0|$ weights ($\ll P^{7/8} \log P$ in total); on the collision band $|wX''| \in [\frac{1}{4}, 4] \lambda_a$, Lemma 3.9 with $(\alpha, \beta) = (\frac{15}{8}, \frac{3}{2})$ and $M \leq 1.5 k|j| P^{-1/8}$ gives, per window and summed with the $O(\log P)$ band mass, $\ll ((k|j|)^{1/2} P^{15/16} + (k|j|)^{-1/2} P^{1/16} \cdot k|j| P^{3/8} + (k|j|)^{1/6} P^{1-3/16} + (k|j|)^{-1/6} P^{1/2+3/16}) \log P \ll (k|j|)^{1/2} P^{15/16} \log P$. The main estimate is Lemma 3.3 per frozen run at scale λ_a : run lengths $\geq \frac{1}{22} P^{1/4} / (|j|+1)$ (Lemma 5.1(iii)), and $\lambda_a^{-1/2} \leq 0.92 (k|j|)^{-1/2} P^{1/16} \ll$ run length, so

$$\sum_{\text{runs}} (\ell \lambda_a^{1/2} + \lambda_a^{-1/2}) \leq 1.3 (k|j|)^{1/2} P^{15/16} + 21 (|j|+1) |j|^{-1/2} k^{-1/2} P^{13/16}.$$

Summed over the eight carry branches and $|j| \leq 3$ with the $O(\log^3 P)$ piece masses: $\ll (k)^{1/2} P^{15/16+\varepsilon} \leq 1.8 P^{1/48} P^{15/16+\varepsilon} = 1.8 P^{23/24+\varepsilon}$ at $k \leq P^{1/24}$ — the absorbed $(k|j|)^{1/2}$ loss, exactly at the bottleneck.

(5b) Zero-offset branches ($j = 0$). Anchor curvature $\lambda_0 \in [0.2, 0.9] k h_1 h_2 P^{-5/8}$ (composite exponent $\frac{11}{8}$), runs of length $\geq \frac{1}{22} P^{3/4} / (h_1 h_2)$. Let $\mu = 0.84 \max(u h_1, u' h_2) P^{-3/4}$ be the strongest differenced-wave scale present. Three regimes.

- *Anchor-dominant* ($60\mu \leq \lambda_0$): Lemma 3.3 per run at λ_0 , all else dominated at margin ≥ 20 : $\leq 1.3 (k h_1 h_2)^{1/2} P^{11/16} + 49 (h_1 h_2 / k)^{1/2} P^{9/16}$.
- *Mode-dominant* ($\mu \geq 60\lambda_0$, i.e. $u h_1 \geq 60 k h_1 h_2 P^{1/8}$ -form): Lemma 5.2(i) with the undifferenced anchor as decoration: its run boundaries number $\leq 22 h_1 h_2 P^{1/4} \leq 22 P^{5/16}$, cost $\leq 22 P^{5/16} \cdot 3.4 (u h_1)^{-1/2} P^{3/8} \leq 75 P^{11/16}$; its smooth part is dominated at margin ≥ 20 by hypothesis; its θ -coefficient $\leq 1.2 k h_1 h_2 P^{3/8} \leq 1.2 P^{1/2}$ produces at most $1.2 P^{1/2}$ windows (boundary cost $\leq 1.2 P^{1/2} \cdot 3.4 (u h_1)^{-1/2} P^{3/8} \leq 4.1 P^{7/8} (u h_1)^{-1/2}$) whose centre modes w are treated exactly as in Lemma 5.2(i), Stage 5. Total $\ll ((u h_1)^{1/2} P^{5/8} + (h_1 / u)^{1/2} P^{7/8} + P^{7/8}) P^\varepsilon$.
- *Middle band* ($\frac{1}{60} \leq \mu / \lambda_0 \leq 60$; coefficient mass $O(1)$ per layer): here up to three curvature scales meet — the u - and u' -modes merge into a single $\nu^{3/4}$ -term with frozen coefficient $\frac{3}{2} (u G_1 + u' G_2)$ per cell intersection (at most $1.5 (h_1 + h_2) P^{1/2} + 2$ cells), the anchor is the $\nu^{11/8}$ -term, and the window modes are $\nu^{3/2}$ -terms. Lemma 3.10 with $(\alpha, \beta, \gamma) = (\frac{3}{4}, \frac{11}{8}, \frac{3}{2})$ and $M \geq 0.2 k h_1 h_2 P^{-5/8} \geq 0.2 P^{-5/8}$ gives, per cell and summed,

$$\ll \left(P M^{1/2} + (h_1 + h_2) P^{1/2} M^{-1/2} + P (M/P)^{1/6} + P^{1/2} (M/P)^{-1/6} + P (M/P^2)^{1/14} + P^{3/4} (M/P^2)^{-1/14} \right) \log P,$$

and with $M \in [0.2P^{-5/8}, 1.35P^{-1/4}]$ every term is $\leq 1.3P^{15/16}$: in order, $\leq 1.2P^{7/8}$; $\leq 2.3(h_1+h_2)P^{1/2+5/16} \leq 5P^{1/24+13/16} \leq 5P^{7/8}$; $\leq P^{1-13/48}$; $\leq 1.3P^{1/2+13/48} \leq 1.3P^{37/48}$; $\leq P^{1-3/16}$; $\leq 1.2P^{3/4+3/16} = 1.2P^{15/16}$.

In all three regimes the $j = 0$ pieces total $\ll P^{15/16+\varepsilon} \ll P^{23/24}$.

Step 6 (assembly). Additive costs: the M_1 deletion ($2.7P^{1/4}$); majorants (≤ 3 sawtooth layers at $4P/J_2 = 4P^{23/24}$ each, plus $8P^{3/4}$, $46P^{3/4}$, and the displayed window residuals $\leq 7P^{7/8}$); flat costs ($\leq 46P^{3/4}$ per layer). Multiplicative costs: mode masses over at most three expansion layers plus the shift devices, $O(\log^3 P) = P^\varepsilon$. Piece totals: Step 4, $\ll P^{23/24+\varepsilon}$; Step 5a, $\leq 1.8P^{23/24+\varepsilon}$; Step 5b, $\ll P^{15/16+\varepsilon}$; slow modes and (D3) remnants, $\ll P^{7/8+\varepsilon}$. Hence $|T_2| \ll P^{23/24+\varepsilon}$, and by Step 1

$$|T_1|^2 \leq 2P^{2-1/24} + CP^{1+23/24+\varepsilon} \Rightarrow |T_1| \ll P^{1-1/48+\varepsilon}, \quad |K_c|^2 \leq 2P^{2-1/48} + CP^{2-1/48+\varepsilon},$$

so $|K_c| \ll P^{1-1/96+\varepsilon}$. Exponents deliberately unoptimized. $\square$

Two remarks on the constants. First, the k -range is sharp for this assembly: the class-(5a) loss $(k|j|)^{1/2}P^{15/16}$ meets the wave-piece bottleneck $P^{23/24}$ exactly at $k = P^{1/24}$, which is why the theorem is uniform on the original range with no k -explicit statement needed. Second, the exponent $\frac{1}{96} = \frac{1}{4} \cdot \frac{1}{24}$ traces entirely to the depth-2 strength $P^{23/24}$ of the exact level-2 waves (Lemma 5.2(ii)); any improvement of the wave bound improves δ proportionally.

Corollary 5.4 (low-exponent families). Fix a finite set $\mathcal{A} \subset (0, \frac{9}{8}] \setminus \{\frac{1}{4}, \frac{3}{4}\}$. The bound of Theorem 5.3 holds, with constants depending on $\mathcal{A}$, for every monomial-pattern family $c^{(r)} \asymp kP^{\alpha-r}$ with $\alpha \in \mathcal{A}$, uniformly for $k \leq P^{1/24}$.

Proof. Every constraint of the proof is monotone in α on $(0, \frac{9}{8}]$: (C1) relaxes to $kh_1h_2 \leq P^{5/4-\alpha}$, and every coefficient size, window drift, and flat cost shrinks. The anchor composites have exponents $\alpha + \frac{3}{4}$ (offset branches) and $\alpha + \frac{1}{4}$ (zero-offset); the single-sign requirement is $(\alpha + \frac{3}{4})(\alpha - \frac{1}{4}) \neq 0$ resp. $(\alpha + \frac{1}{4})(\alpha - \frac{3}{4}) \neq 0$, i.e. $\alpha \notin \{\frac{1}{4}, \frac{3}{4}\}$. Where the window-centre composite margin degenerates (the analogue of $\frac{945}{512} - \frac{27}{64}$ can vanish at one interior $\alpha_0 = (\sqrt{10}-1)/4$), the collision is covered by Lemma 3.9 with the pair $(\alpha + \frac{3}{4}, \frac{3}{2})$, valid since $\alpha + \frac{3}{4} \neq \frac{3}{2}$ on $\mathcal{A}$; Lemma 3.10 runs with $(\frac{3}{4}, \alpha + \frac{1}{4}, \frac{3}{2})$, pairwise distinct for $\alpha \notin \{\frac{1}{4}, \frac{3}{4}\}$. The exponent sets E of Lemmas 3.9–3.10 are enlarged to include $\alpha + \frac{3}{4}$, $\alpha + \frac{1}{4}$ for $\alpha \in \mathcal{A}$, a finite set. $\square$

6. Applications: depth four complete, and deeper contracting words

Theorem 6.1 (the OOO^* splits; depth four complete). For $w \in \{OOOE, OOOO\}$,

$$\#\{n \leq N \text{ odd} : \text{word}_4(n) = w\} = \frac{N}{16} + O(N^{1-1/96+\varepsilon}).$$

Hence **every** itinerary word class of depth at most four satisfies $\#\{n \leq N\} = 2^{-|w|}N + O(N^{1-\delta_w})$ with explicit $\delta_w > 0$: depth-4 parity equidistribution is complete.

Proof. The class indicator expands (Lemma 3.6, then Lemma 3.5 at truncation $J_3 = P^{1/96}$, error $P/J_3 = P^{1-1/96}$) into mode sums $S_{ijk} = \sum_n e(\frac{i}{2}X + \frac{j}{2}Y + \frac{k}{2}v^{3/2})$ over dyadic blocks. Three applications of the Taylor pattern of Lemma 4.3(i), taken to second order, give the exact identity (odd $n \geq 5$)

$$v^{3/2} = -\frac{5}{64}n^{27/8} + \frac{9}{32}mn^{15/8} - \frac{45}{64}m^2n^{3/8} + \frac{15}{64}vn^{9/8} + \frac{45}{32}vmn^{-3/8} - \frac{9}{64}vm^2n^{-15/8} + \text{err},$$

$|\text{err}| \leq \frac{3}{4}n^{-9/8}$: the fourth-letter phase is a polynomial of degree $(2, 1)$ in the integer pair (m, v) with smooth coefficients. Its v -linear part is vW with W in the family of Theorem 5.3; the vm - and vm^2 -cross terms reduce to the same family by $\xi vm = \xi Xv - \xi \theta v$ and $\xi_2 vm^2 = \xi_2 X^2v - 2\xi_2 X\theta v + \xi_2 \theta^2 v$,

whose θ -cross coefficients are sub-unit and whose smooth v -coefficients $\xi X, \xi_2 X^2 \asymp kP^{9/8}$ are family members. Writing $v = Y - \theta_2$ splits each family term into a smooth-in- m part and a kernel factor $e(-c\theta_2)$. The double differencing of Theorem 5.3 applied to the whole mode phase carries the passengers along: the pure- m polynomial phases difference into the anchor classes of Step 5 (their $(\Delta\Delta\mu)m$ content is sub-unit under (C1), the crosses are smooth per cell, and $\mu\Delta\Delta m$ produces the bounded-offset branch phases with curvature $\asymp k|j|P^{-1/8}$, single-signed by the same monomial-exponent check); the $\frac{i}{2}X$ - and $\frac{i}{2}Y$ -passengers difference to curvatures $\ll ih_1h_2P^{-5/2}$ and $\asymp jh_1h_2P^{-5/4}$, each smaller than every retained curvature scale of Step 5 by a fixed power of P , and the differenced $\frac{i}{2}Y$ -passenger's wave content is exactly a (D1) decoration; and the kernel factor is handled by Steps 2–6 of Theorem 5.3 without change, since those steps use only the derivative pattern of c , not its specific form. Assembly and summation over $k \leq J_3$ with $1/k$ -weights give $\sum_k \frac{1}{k} P^{1-1/96} + P^{1-1/96} \ll P^{1-1/96+\varepsilon}$; dyadic blocks sum to $N^{1-1/96+\varepsilon}$. $\square$

The certified-descent density stays $13/16$ at four steps — $OOO*$ is non-contracting at depth 4 ($3^3 > 2^4$) — but the kernel theorem opens the contracting words at depths five, seven, and eight.

Theorem 6.2 (the length-5 contracting splits).

$$\#OOOEE(N), \#OOOEO(N) = \frac{N}{32} + O(N^{1-1/96+\varepsilon}), \quad \#OOEOE(N), \#OOEOO(N) = \frac{N}{32} + O(N^{43/48+\varepsilon}).$$

Proof. $OOO*$. The class indicator is the Theorem-6.1 indicator times $\frac{1}{2}(1 \pm \psi(z^{1/2}))$, $z = J^3(n)$: branch-consistent by Lemma 3.6, since on the $OOOE$ cylinder z is even, so the fifth letter is the even-branch value. Vaaler-expand the fifth wave at truncation $P^{1/24}$. The Lemma 4.3(i) pattern along the chain replaces $\frac{l}{2}z^{1/2}$ by $\frac{l}{2}n^{27/16} - \frac{9l}{16}n^{3/16}\theta$ at absorbable decaying cost. The θ -sawtooth has coefficient $\asymp lP^{3/16} < P$; its shifted-window expansion (drift-1 length $P^{13/16}/l$, window $T = P^{1/8} \ll lP^{3/16}$) produces X -modes of size $\asymp lP^{3/16}$ — ordinary first-letter passengers within Theorem 6.1's budget — and the smooth chirp $e(\frac{l}{2}n^{27/16})$ has curvature $\asymp lP^{-5/16}$, subdominant to every retained scale. Theorem 6.1 applies with these passengers.

$OOEO*$. On the $OOEO$ cylinder the fifth-letter phase is $\frac{k}{2}w^{3/2}$ with $w = J^3(n)$ odd. Exact linearization writes it as $\frac{k}{2}n^{27/16} - C\theta - B\theta_w$ with $B = \frac{3k}{4}v^{1/4} \asymp kn^{9/16}$, $C = \frac{9k}{16}n^{3/16}$, and $\theta_w = \{v^{1/2}\}$. Expand $e(-B\theta_w)$ on drift-1 intervals of length $P^{7/16}/k$ (there are $\asymp kP^{9/16}$ of them; window $T_w = P^{1/4} \ll kP^{9/16}$, majorant $P^{3/4}$). At frequency $-B + t$, $|t| \leq T_w$, the combined phase is $-\frac{k}{4}v^{3/4} + tv^{1/2}$ plus passengers; linearizing both powers, the θ -coefficients cancel at the window centre up to a residual $\asymp kn^{3/16}$, leaving one slow θ -sawtooth, expanded on the same intervals (window $P^{1/8}$, majorant $P^{7/8}$). The resulting smooth phase has curvature $\lambda_2 = (-\frac{297}{1024} + \frac{27}{128})kn^{-5/16}(1 + o(1)) \asymp kn^{-5/16}$, single-signed since the two rational coefficients do not cancel. Lemma 3.3 on each interval gives $\ll k^{-1/2}P^{9/32}$; over $\asymp kP^{9/16}$ intervals, $S_k \ll k^{1/2}P^{27/32+\varepsilon}$. Balancing $J_5^{1/2}P^{27/32} = P/J_5$ at $J_5 = P^{5/48}$ gives $P^{43/48}$; dyadic blocks sum to $N^{43/48+\varepsilon}$. $\square$

Theorem 6.3 (the length-7 engine splits). All eight classes

$$\#OOEOO**(N), \#OOOEO**(N) = \frac{N}{128} + O(N^{1-1/96+\varepsilon}).$$

Proof. $OOEOO**$. With $w = J^3(n)$ odd, $p = \lfloor w^{3/2} \rfloor$, $\theta_p = \{w^{3/2}\}$, the exact rearrangement

$$p^{3/2} = -\frac{5}{4}v^{9/8} + \frac{9}{4}wv^{5/8} - \frac{3}{2}w^{3/4}\theta_p + E, \quad 0 \leq E \leq \frac{3}{8}p^{-1/2} + \frac{45}{32}v^{1/8},$$

obtained from two Taylor steps and the identity $\theta_w = v^{1/2} - w$, eliminates the supercritical naive coefficient $\frac{9}{4}v^{5/8} \asymp n^{45/32} > n$ of θ_w ; the remaining sawtooth θ_p has coefficient $\asymp n^{27/32} < n$

and derivative < 1 . Expanding the smooth powers of v by the linearization pattern: the first-letter content is a chirp of curvature $\asymp n^{17/32}$ (Lemma 3.3); the leading θ_2 -content has coefficient $\asymp n^{33/32}$ — a family of Corollary 5.4 at $\alpha = \frac{33}{32} < \frac{9}{8}$; the remaining θ_2 -amplitudes are $O(n^{9/32})$, inside the engine. The integer block $e(\xi w)$ with $\xi \asymp n^{45/32}$ is handled on frozen gap runs: on the OOEO cylinder $U = v^{1/2}$ has $U'' \asymp P^{-7/8} < 1$, so $\lfloor \Delta U \rfloor$ is constant on runs of length $\asymp P^{7/8}/h$, and per run $\Delta w = J_U + \kappa_w$ with J_U frozen, $e(\xi J_U)$ a smooth chirp, and κ_w an indicator weight — the Lemma 5.1(iii) pattern one level up. The sawtooth θ_p expands on drift-1 intervals of length $P^{5/32}$ with window $T \ll P^{27/32}$, producing X -modes within Theorem 6.2's budget. The seventh letter is the even-branch square root of $q = \lfloor p^{3/2} \rfloor$, with decaying amplitudes only. Theorem 6.2 applies as a passenger theorem.

*OOEO**.* The analogous rearrangement at base $z^{1/2}$ ($s = \lfloor z^{1/2} \rfloor$: $s^{3/2} = -\frac{1}{2}z^{3/4} + \frac{3}{2}s z^{1/4} + E$) plus the $z^{3/4} \rightarrow n^{81/32}$ linearization chain gives the same four classes of terms — chirp, Corollary-5.4 family at $\alpha = \frac{33}{32}$, engine sawtooths, and an integer- s block on $\lfloor \Delta z^{1/2} \rfloor$ -runs of length $\asymp P^{5/16}$. Same classes of terms. In both cases the binding budget is the Corollary-5.4 family (and, for the *OOEO*-prefix, the parent kernel budget of Theorem 6.1), i.e. $N^{1-1/96+\varepsilon}$; the engine's own $N^{43/48}$ sits below it. $\square$

Theorem 6.4 (the length-8 engine quartet). The four contracting length-8 classes satisfy

$$\#OOEOEOE(N), \#OOEOOOEE(N), \#OOEOEOEOE(N), \#OOEOOOEE(N) = \frac{N}{256} + O(N^{1-1/96+\varepsilon}).$$

Proof. These are exactly the contracting length-8 children of the classes counted by Theorem 6.3 (appending E to the four five-odd words; the six-odd words need length ≥ 10). Write $x_1 = n$ and $x_{t+1} = \lfloor x_t^{3/2} \rfloor$ or $\lfloor x_t^{1/2} \rfloor$ as the letters dictate; the eighth-letter wave has phase argument X_7 , the real number with $x_8 = \lfloor X_7 \rfloor$. On each of the four parents, applying the two-term Taylor identity with one-signed remainder at every one of the seven levels gives the exact chain

$$X_7 = n^{243/128} - \sum_i B_i(n) \theta_i + E, \quad |E| < 1 \quad (n \geq 51),$$

with $\theta_i \in [0, 1)$ the level- i floor defect and every coefficient subcritical: the complete inventory of growing coefficients over the four words is $\frac{27}{16}x_3^{11/32} \asymp n^{99/128}$, $\frac{3}{2}x_6^{1/4} \asymp n^{81/128}$, $\frac{81}{64}n^{51/128}$, and $\frac{9}{8}x_4^{3/16}$ resp. $\frac{9}{8}x_4^{1/16} \asymp n^{27/128}$; all other coefficients decay. The interleaved even letters keep every intermediate state below the scale $n^{9/4}$ at which an odd letter would produce a supercritical (kernel-class) coefficient — no letter of the quartet sees a kernel. Every sawtooth has coefficient drift $\leq n^{-29/128} < 1$: expand each on drift-1 windows with budget $T = P^{1/16}$ (majorant $P^{15/16}$); the window-centre values recombine with the smooth chirp $(k/2)n^{243/128}$, and the residual modes carry curvature $\leq Tn^{-7/8}$, subdominant to the chirp curvature $\lambda \asymp kn^{-13/128}$. On the intersected windows (length $\geq n^{29/128} \gg \lambda^{-1/2}$) Lemma 3.3 gives $\ll k^{1/2}n^{45/256}$ per window, hence $S_k \ll k^{1/2}P^{243/256}$; balancing the Vaaler majorant P/J_8 at $J_8 = P^{13/384}$ gives $P^{1-13/384}$ for the eighth-letter wave. Letters one through seven ride as passengers with the budgets of Theorems 6.1–6.3, and the parent budget $N^{1-1/96+\varepsilon}$ dominates. $\square$

Corollary 6.5 (certified-descent densities 7/8, 57/64, and 29/32). The class of starts carrying a uniform power-envelope descent certificate of length at most five — evens, *OE*, *OOEE*, *OOOEE*, *OOEOE* — has natural density 7/8; adding the length-7 words *OOEOOOEE* and *OOEOEOEE* raises the density of the length- ≤ 7 certified class to 57/64; adding the length-8 quartet of Theorem 6.4 raises the length- ≤ 8 certified class to 29/32.

Proof. Densities $\frac{1}{2} + \frac{1}{4} + \frac{1}{16} + \frac{1}{32} + \frac{1}{32} = \frac{7}{8}$, $\frac{7}{8} + \frac{1}{128} + \frac{1}{128} = \frac{57}{64}$, and $\frac{57}{64} + \frac{4}{256} = \frac{29}{32}$. The cylinders are disjoint, and each new cylinder is counted with a power saving by Theorems 6.2–6.4. Contraction is Proposition 3.1: $3^3 = 27 < 32 = 2^5$ on the length-5 words, $3^4 = 81 < 128 = 2^7$ on the length-7 words, and $3^5 = 243 < 256 = 2^8$ on the length-8 words. $\square$

The leftover $\frac{3}{32}$ decomposes exactly: the $OOOO*$ tree ($\frac{1}{16}$, blocked at its root by the level-3 kernel of Section 7, first contracting word $OOOOEEE$), the expanding O -children of the quartet splits ($\frac{4}{256}$), and the two six-odd trees $OOEOOOO*$, $OOOEEOOO*$ ($\frac{4}{256}$). The split is structural: an odd letter applied at state scale n^σ produces a letter-phase coefficient $\asymp n^{\sigma/2}$, the engine and Corollary 5.4 cover $\sigma \leq 9/4$, and even letters halve σ while odd letters multiply it by $3/2$. The non- $OOOO$ leftover therefore keeps thinning at every depth by further engine theorems with diminishing increments, while the $OOOO$ tree is monolithically blocked by the level-3 kernel.

As with Corollary 4.2, these are densities of uniform certificate classes. They are not densities of all descent certificates and not densities of starts that reach 1.

7. The Terras-style reduction and the frontier

The depth-by-depth counting assembles into a conditional Terras-style statement, and the reduction is unconditional:

Proposition 7.1 (equidistribution implies density-one descent). Let $d \geq 1$ and suppose that for every itinerary word w of length d (over all starts, first letter the parity of n),

$$|\#\{n \leq N : \text{word}_d(n) = w\} - 2^{-d}N| \leq E_d(N).$$

Then the starts with no contracting prefix of length $\leq d$ number at most

$$e^{-cd}N + 2^d E_d(N), \quad c = 2\left(\frac{\log 2}{\log 3} - \frac{1}{2}\right)^2 > 0.0342.$$

Every other start $n \geq 2$ satisfies $J^t(n) < n$ for some $t \leq d$ with the uniform power-envelope certificate of Proposition 3.1. Consequently, if $E_d(N) = O_d(N^{1-\delta_d})$ with $\delta_d > 0$ for every d , the set of starts admitting a finite descent certificate has natural density 1.

Proof. A word w of length d has a contracting prefix iff $3^{o_t} < 2^t$ for some $t \leq d$, where o_t counts odd letters among the first t . If w has no contracting prefix then $3^{o_d} \geq 2^d$, i.e. $o_d \geq \beta d$ with $\beta = \log 2 / \log 3 = 0.6309 \dots$. The number of such words is at most $2^d \Pr[\text{Bin}(d, \frac{1}{2}) \geq \beta d] \leq 2^d e^{-2(\beta-1/2)^2 d}$ by Hoeffding's inequality. Each word class has at most $2^{-d}N + E_d(N)$ members; summing over the non-contracting words gives the count. The density-one statement follows by letting $d \rightarrow \infty$ slowly with N (any $d(N) \rightarrow \infty$ with $2^d E_d(N)/N \rightarrow 0$). $\square$

Every E-rooted word has a contracting prefix at length one ($3^0 < 2$), so the hypothesis only ever consumes O-rooted class bounds. Sections 4–6 prove it *unconditionally at every depth* $d \leq 4$, so the base cases of Proposition 7.1 are theorems. The first open case is depth 5: the $OOOO*$ split. It has an exact shape, one nesting deeper than Theorem 5.3. Write $Z = v^{3/2}$, $z = \lfloor Z \rfloor$, $\theta_3 = Z - z$; after four odd letters the fifth is the parity of $\lfloor z^{3/2} \rfloor$.

Lemma 7.2 (level-3 kernel reformulation). For odd $n \geq 5$,

$$\frac{1}{2}(v^{9/4} - z^{3/2}) - \frac{3}{4}z^{1/2}\theta_3 = R_3, \quad 0 \leq R_3 \leq \frac{3}{16}z^{-1/2}.$$

Consequently the fifth-letter phase is, up to $kR_3 \ll kn^{-27/16}$ per term, the exponential sum of the level-3 local floor defect, with coefficient $c = \frac{3k}{4}z^{1/2} \asymp kn^{27/16}$: Lemma 5.1(i) with (m, v) replaced by (v, z) .

Proof. Taylor of $(z + \theta_3)^{3/2}$ at z , with $Z^{3/2} = v^{9/4}$. $\square$

For smooth c with $c \asymp kP^{27/16}$ and $c' \asymp kP^{11/16}$ on $n \sim P$ (the $z^{1/2}$ -shaped family), define

$$K_3(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e(c(n) \{ \lfloor \lfloor n^{3/2} \rfloor^{3/2} \}).$$

Conjecture 7.3 (level-3 kernel cancellation). $K_3(P) \ll P^{1-\delta}$ for some $\delta > 0$, uniformly over the family above with $k \leq P^\varepsilon$.

The boundary between Theorem 5.3 and Conjecture 7.3 is quantitative and sharper than a derivative count. The smooth model iterates: where Theorem 5.3 used two Weyl differencings against $Y'' \asymp P^{1/4} \gg 1 > P^{-3/4} \asymp Y'''$, the level-3 model $n^{27/8}$ has $G''' \asymp P^{3/8} \gg 1 > P^{-5/8} \asymp G^{(4)}$ and predicts three. But the prediction does not descend to the nested floors. The kernel weight now has $c' \asymp kP^{11/16} \gg 1$, so no drift-1 interval exists for any expansion of the weight itself; the branch decomposition of Lemma 5.1(iii) has no analogue at the v -level, because v jumps by $\asymp n^{5/4}$ per step and the branch set is a *product* of two carry lattices, not a copy of one; and the forced inner linearization $v^{3/2} = m^{9/4} - \frac{3}{2}m^{3/4}\theta_2 + E_2$ trades θ_3 for a sawtooth family at coefficient scale $kn^{45/16}$, above the $9/4$ threshold where every method of this paper stops.

What survives at the frontier is a model problem and one theorem about it. Stripping every problem-specific structure (carries, defects, nesting) from K_3 leaves the *amplitude-product* sums

$$S = \sum_{t \leq L} e(A(t) \{B(t)\}), \quad 1 \ll A' \ll A,$$

with smooth monomial-type A, B (the instance above has $A \asymp P^{27/16}$, $A' \asymp P^{11/16}$). For $A' \ll 1$ partial summation makes the amplitude a tame passenger and the classical single-floor machinery applies; for $A' \gg 1$ we know of no nontrivial deterministic bound by any method. The generic statement, however, is a theorem — and its proof uses no harmonic analysis at all:

Theorem 7.4 (shift-averaged square-root cancellation). Let $A_1 < \dots < A_L$ be reals with $|A_t - A_{t'}| \geq A'_{\min}|t - t'|$ for some $A'_{\min} \geq 1$, and let $x_1, \dots, x_L$ be arbitrary reals. For $S_\lambda = \sum_{t \leq L} e(A_t \{x_t + \lambda\})$,

$$\left| \int_0^1 |S_\lambda|^2 d\lambda - L \right| \leq \frac{6}{\pi} \frac{L}{A'_{\min}} (\log L + 1).$$

In particular $|S_\lambda| \leq \sqrt{L/\varepsilon}$ outside a shift set of measure $\varepsilon(1 + o(1))$: two-sided square-root cancellation for almost every shift of the fractional argument, regardless of the sequence B .

Proof. Expand the square; the diagonal gives L . For $t \neq t'$, the function $\varphi(\lambda) = A_t \{x_t + \lambda\} - A_{t'} \{x_{t'} + \lambda\}$ is piecewise linear on $[0, 1)$ with *real* slope $A_t - A_{t'}$ on at most three arcs (jumps at $1 - \{x_t\}$ and $1 - \{x_{t'}\}$); on each arc $|\int e(\varphi) d\lambda| \leq 1/(\pi|A_t - A_{t'}|)$. Summing, $\sum_{t \neq t'} 3/(\pi A'_{\min}|t - t'|) \leq (6/\pi)(L/A'_{\min})(\log L + 1)$. $\square$

The mechanism deserves note: the amplitude separation $A' \gg 1$ — the very property that defeats every character expansion, since a Fourier window centred at the amplitude drifts by A' harmonics per step — is exactly what makes the shift average trivial. What remains of Conjecture 7.3 after

Theorem 7.4 is a de-randomization: the deterministic sum is the single shift $\lambda = 0$ in a family that cancels almost surely.

Conjecture 7.5 (the pure amplitude-product model). $|S| \leq L^{1-\delta}$ for some $\delta > 0$, for smooth monomial-type A, B with $1 \ll A' \ll A$ as above.

Three structural facts locate the difficulty of the de-randomization. No second averaging variable exists: amplitude separation forces any two sample points with $|A(p) - A(q)| \leq 1$ to coincide, and every family average available in the application re-enters either the differenced-kernel class or the amplitude-product class itself. The inverse theory is self-similar: any concentration or discrepancy inverse for $A\{B\} \bmod 1$ is a statement about $\sum e(jA\{B\})$ — the same class with amplitude jA — so the class is closed under its own inverse theorems and no bootstrapping is possible. And the metric statement does not transfer: $|dS_\lambda/d\lambda| \leq 2\pi A_{\max} L$ almost everywhere, so S_λ decorrelates at shift scale $1/A_{\max}$, and an almost-all- λ theorem leaves $\asymp \varepsilon A_{\max}$ exceptional cells among $\asymp A_{\max}$ — no measure argument pins $\lambda = 0$. The deterministic content of Conjecture 7.5 is a specific-point-in-metric-theory problem, the same species as the normality of $\sqrt{2}$: a class whose known successes all use special arithmetic.

Two analytic shortcuts around Conjecture 7.3 fail at recorded points, and we state them once: composing gap cells across two levels fails because on a cell where the first-level gap is constant the second-level gap still takes a new value at essentially every point, so no usable sub-cell survives; and reindexing by the image m strips one nesting level but introduces a fiber indicator of density $\asymp m^{-1/3}$, whose sawtooth requires savings beyond the exponent $1/3$ while the engine saves only $1/24$. A fuller account of parked routes lives in the companion repository, not here.

The open question, stated once:

It is open whether almost every odd-to-odd start has a finite descent certificate. By Proposition 7.1 that would follow from all-depth parity equidistribution, which is now a theorem through depth four; the first open case is the *OOOO** kernel of Conjecture 7.3, whose model form cancels for almost every shift (Theorem 7.4) and whose deterministic instance is Conjecture 7.5.

8. Software note

A repository accompanies the paper: https://github.com/sneakyweasel/balanced_ternary/. It is not required to read or check any proof above. It contains Lean formalizations of the exact floor identities cited in Section 1.1 and of the companion manuscript’s finite-word theorems, scaled-integer validations of the linearization identities of Lemmas 4.3, 4.6, 5.1, and 7.2 and of the eighth-letter chain of Theorem 6.4, and exact-phase numerical probes of the sums K_c , K_3 , and the differenced sums of Lemma 5.2 and Theorem 5.3, all of which exhibit cancellation at square-root scale — stronger than the theorems claim and stronger than they need. Probes and validations are checks, not proofs, and no statement in this paper depends on them.

Acknowledgments

I used large language models extensively while drafting and revising the text, organizing companion notes, and as an interactive assistant for Lean statements, tests, and literature records. The models are not authors. The theorems of this paper are human proofs from the cited classical inequalities; the Lean certificates cover only the exact floor identities and the companion’s finite-word theorems. I take full responsibility for the contents.

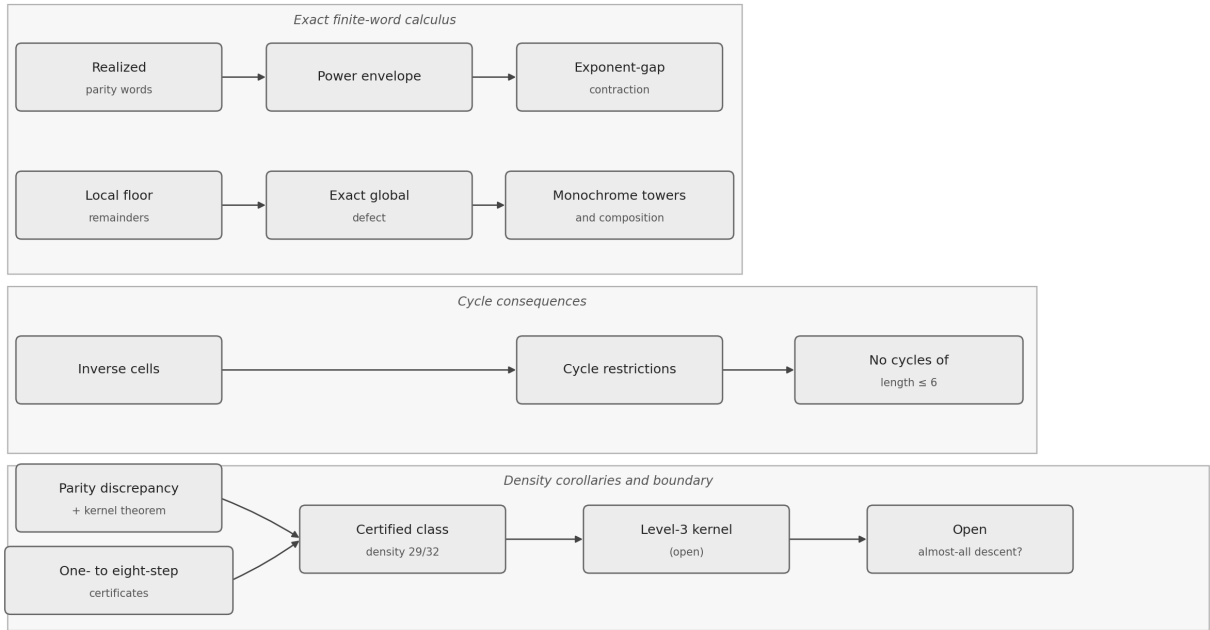


Figure 1: The theorem flow of the paper. The exact finite-word calculus of the companion manuscript feeds the contraction certificates; the discrepancy calculus with the kernel theorem counts every itinerary class through depth four and the contracting words of lengths five through eight (certified density $29/32$), leaving the level-3 kernel — generically cancelling by the shift-average theorem — and with it almost-all descent, open.

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